CHAPTER 04

# Analyzing Organic Reactions

## Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 561ac9387f96a3109b02d84c092c477affb477da6b3bffbc8c140c43f9550841

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

### Coverage checklist

ch04:exercise:pka-equilibrium | pka-equilibrium:stage-1, pka-equilibrium:stage-2, pka-equilibrium:stage-3

ch04:exercise:identify-partners | identify-partners:stage-1, identify-partners:stage-2, identify-partners:stage-3

ch04:exercise:sn2-rate | sn2-rate:stage-1, sn2-rate:stage-2, sn2-rate:stage-3

ch04:exercise:carbon-redox | carbon-redox:stage-1, carbon-redox:stage-2, carbon-redox:stage-3

ch04:exercise:selective-reduction | selective-reduction:stage-1, selective-reduction:stage-2, selective-reduction:stage-3

ch04:lesson:acids | acids:rule, acids:cue, acids:trap, acids:app, acids:step-1, acids:step-2, acids:step-3

ch04:lesson:partners | partners:rule, partners:cue, partners:trap, partners:app, partners:step-1, partners:step-2

ch04:lesson:substitution | substitution:rule, substitution:cue, substitution:trap, substitution:app, substitution:step-1, substitution:step-2

ch04:lesson:redox | redox:rule, redox:cue, redox:trap, redox:app, redox:step-1, redox:step-2, redox:step-3

ch04:lesson:selectivity | selectivity:rule, selectivity:cue, selectivity:trap, selectivity:app, selectivity:step-1, selectivity:step-2

ch04:molecule:aceticacid | aceticacid

ch04:molecule:acetate | acetate

ch04:molecule:acetaldehyde | acetaldehyde

ch04:molecule:bromoethane | bromoethane

ch04:molecule:tbutylbromide | tbutylbromide

ch04:molecule:ethanol | ethanol

ch04:molecule:acetone | acetone

ch04:molecule:ketoester | ketoester

ch04:curriculum | scope-1, scope-2, scope-3, scope-4, scope-5

# Curriculum scope

scope-1: 4.1 Acids and Bases

scope-2: 4.2 Nucleophiles, Electrophiles, and Leaving Groups

scope-3: 4.3 Oxidation–Reduction Reactions

scope-4: 4.4 Chemoselectivity

scope-5: 4.5 Steps to Problem Solving

Objective: Estimate acid–base equilibria

Objective: Distinguish SN1 and SN2

Objective: Track oxidation and chemoselectivity

Supplied Kaplan curriculum: Chapter 4; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter04b.xhtml

SN1 stereochemical qualifications: OpenStax 11.4 | https://openstax.org/books/organic-chemistry/pages/11-4-the-sn1-reaction

OpenStax: SN2 nucleophile and solvent effects | https://openstax.org/books/organic-chemistry/pages/11-3-characteristics-of-the-sn2-reaction

OpenStax: aldehyde oxidation conditions | https://openstax.org/books/organic-chemistry/pages/19-3-oxidation-of-aldehydes-and-ketones

OpenStax: amide reduction to amines | https://openstax.org/books/organic-chemistry/pages/21-7-chemistry-of-amides

OpenStax: acyl substitution and conditional reactivity order | https://openstax.org/books/organic-chemistry/pages/21-2-nucleophilic-acyl-substitution-reactions

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

# acids | Acid–base reasoning

Equilibrium favors the weaker acid/base pair

## acids:rule | Core Rules

Brønsted acids donate H+ and bases accept it. Lewis acids accept an electron pair; Lewis bases donate one. In a coordinate covalent bond, both shared electrons originate from the donor; the resulting bond is still covalent. For HA + B ⇌ A− + BH+, K ≈ 10^[pKa(BH+) − pKa(HA)] when the pKa values use comparable conditions. Amphoteric species can act as acids or bases, depending on the partner. Water can accept H+ to form H3O+ or donate H+ to form OH−; bicarbonate can likewise gain or lose a proton.

Schematic summary: Equilibrium favors the weaker acid/base pair

Chemistry diagram: Illustrative acid/conjugate-base pair. Protonation depends on pH; the panels are not separate resonance contributors.

Identify roles — Brønsted H+ transfer or Lewis electron-pair transfer

Compare conjugate pairs — Use compatible conditions

Equilibrium — Favors the weaker acid/base pair

## acids:cue | Memory Cues

Lower pKa means stronger acid. Stabilize the conjugate base to increase acidity. Across a comparable row, electronegativity can stabilize negative charge; down a group, weaker H–X bonds and charge dispersion can dominate. Compare like structures and conditions, rather than applying a single trend everywhere.

Schematic summary: Choose the case that applies

Lower pKa — Stronger acid

More stable conjugate base — Generally stronger acid

Periodic trend — Compare like structures and account for solvent

## acids:trap | Common Mistakes / MCAT Traps

Basicity is an equilibrium property; nucleophilicity is a reaction-rate property. Do not use a single periodic trend across different solvents or unrelated functional groups.

Schematic summary: Shortcut versus a better check

Avoid — Equating basicity and nucleophilicity

Check — Equilibrium strength versus reaction-rate behavior

## acids:app | MCAT Applications

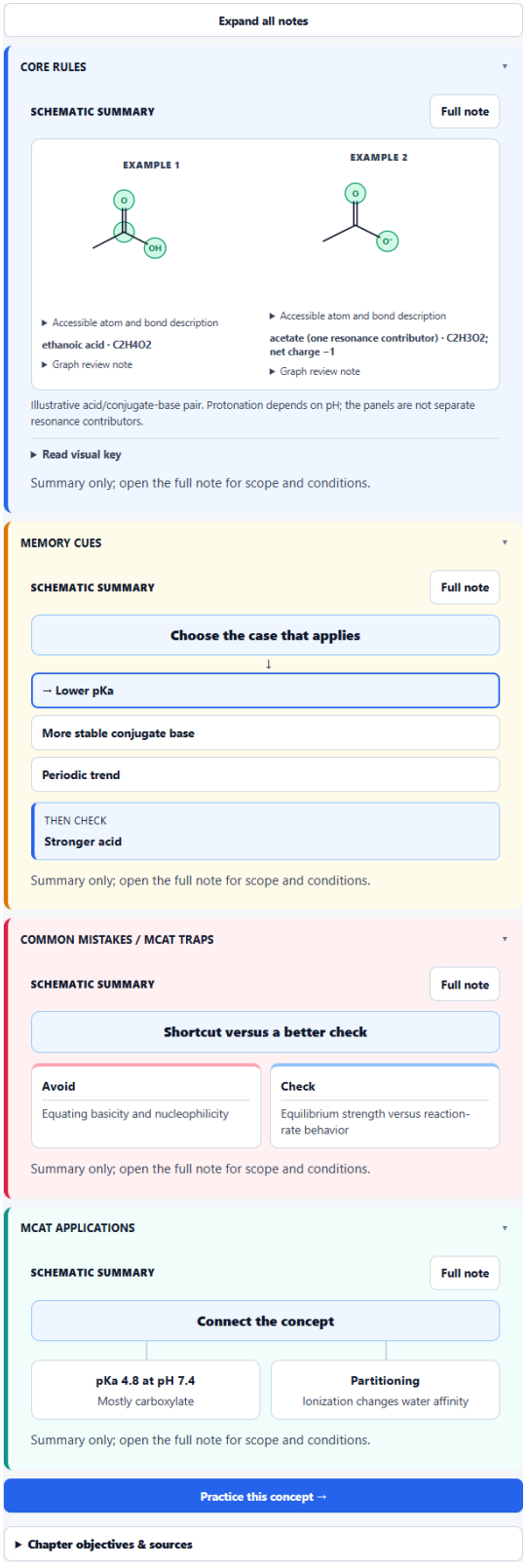
A carboxylic acid with pKa 4.8 is mostly carboxylate at pH 7.4; that changes water solubility and membrane partitioning.

Schematic summary: Connect the concept

pKa 4.8 at pH 7.4 — Mostly carboxylate

Partitioning — Ionization changes water affinity

Screenshot: acids-visuals



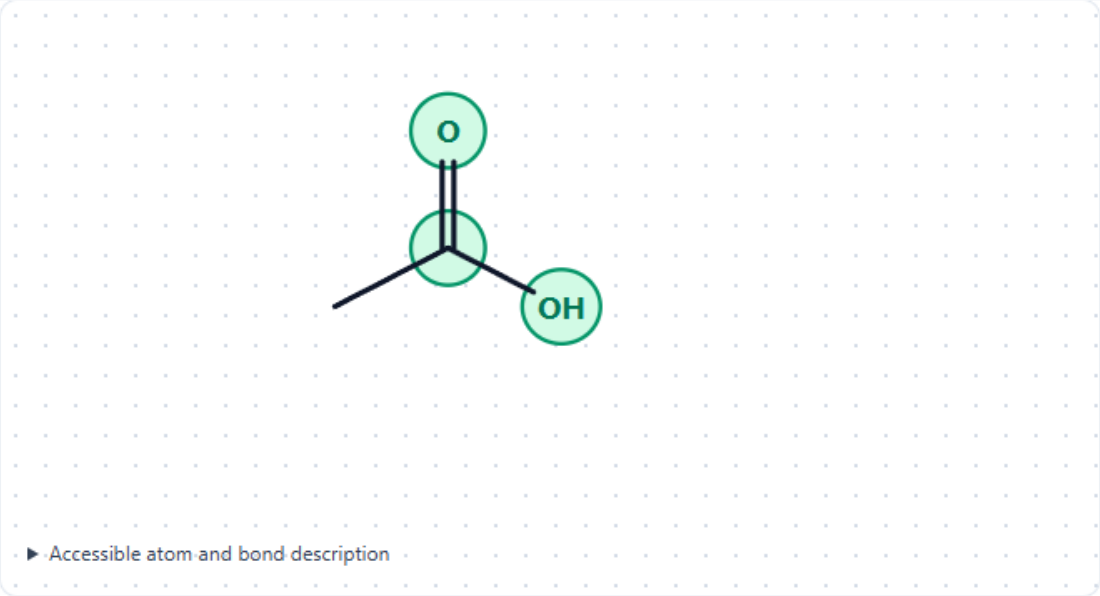

## acids:step-1 | 1. Locate the acidic group

Removing the carboxyl O–H proton gives a carboxylate whose negative charge is shared across two oxygens.

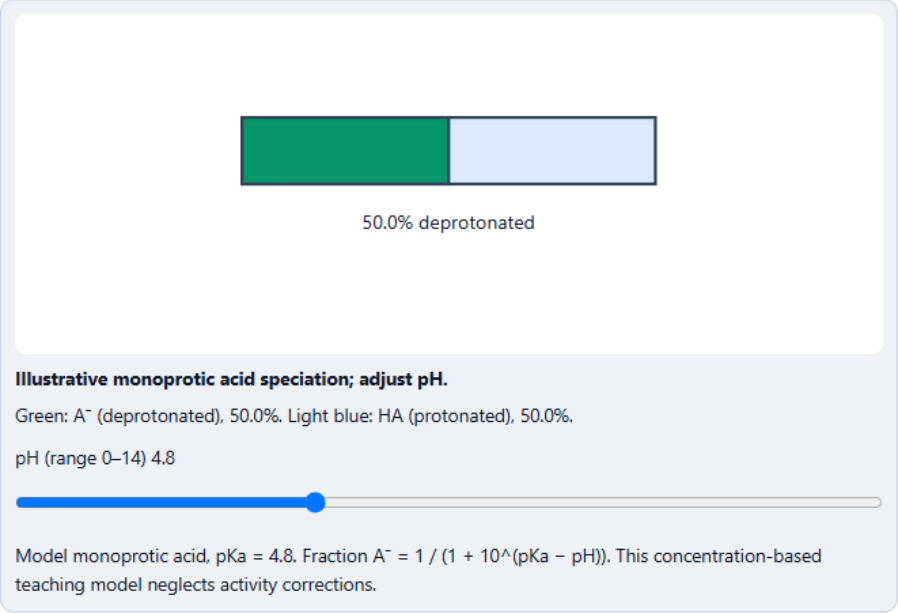
Displayed molecule ID: aceticacid

{"highlights": {"prin": [1, 2, 3]}, "revealName": true}

Screenshot: acids-step-1




Screenshot: acids-step-1-diagram




{"diagramSpecification": {"type": "acid", "pKa": 4.8, "caption": "Illustrative monoprotic acid speciation; adjust pH."}}

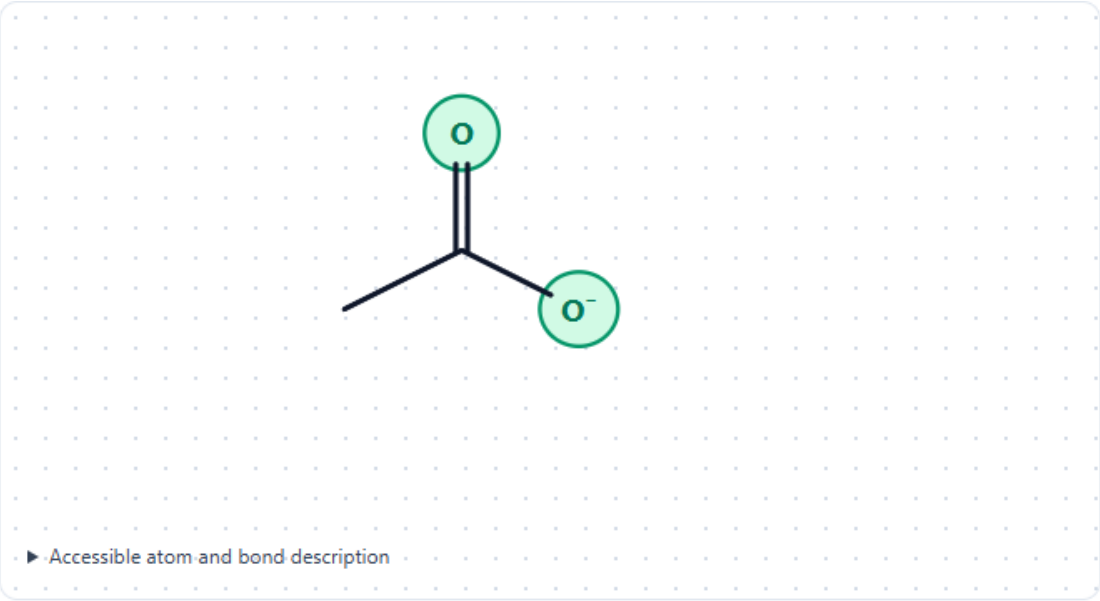
## acids:step-2 | 2. Compare conjugate bases

An alcohol-derived alkoxide lacks the equivalent two-oxygen resonance of a carboxylate. This is a major reason simple carboxylic acids are more acidic than alcohols.

Displayed molecule ID: acetate

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Screenshot: acids-step-2



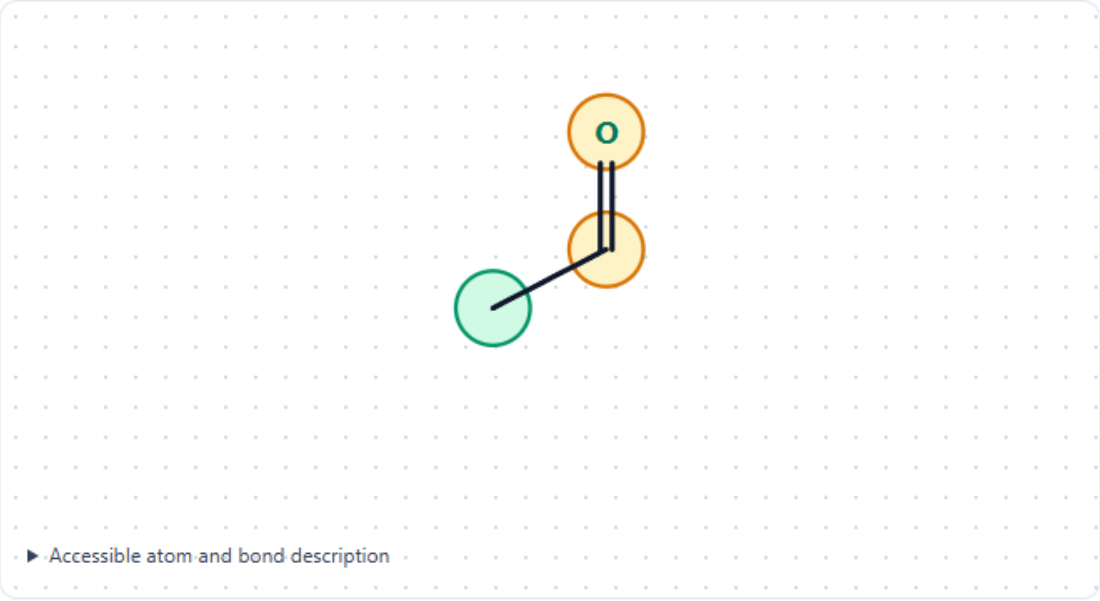

## acids:step-3 | 3. Find alpha hydrogens

In ethanal, the methyl carbon is alpha to the carbonyl. Removing one of its hydrogens with a suitable base can form a resonance-stabilized enolate. Distinguish the acidic alpha C–H sites from the electrophilic carbonyl carbon; an enolate can act as a nucleophile. These hydrogens are implicit in the drawing.

Displayed molecule ID: acetaldehyde

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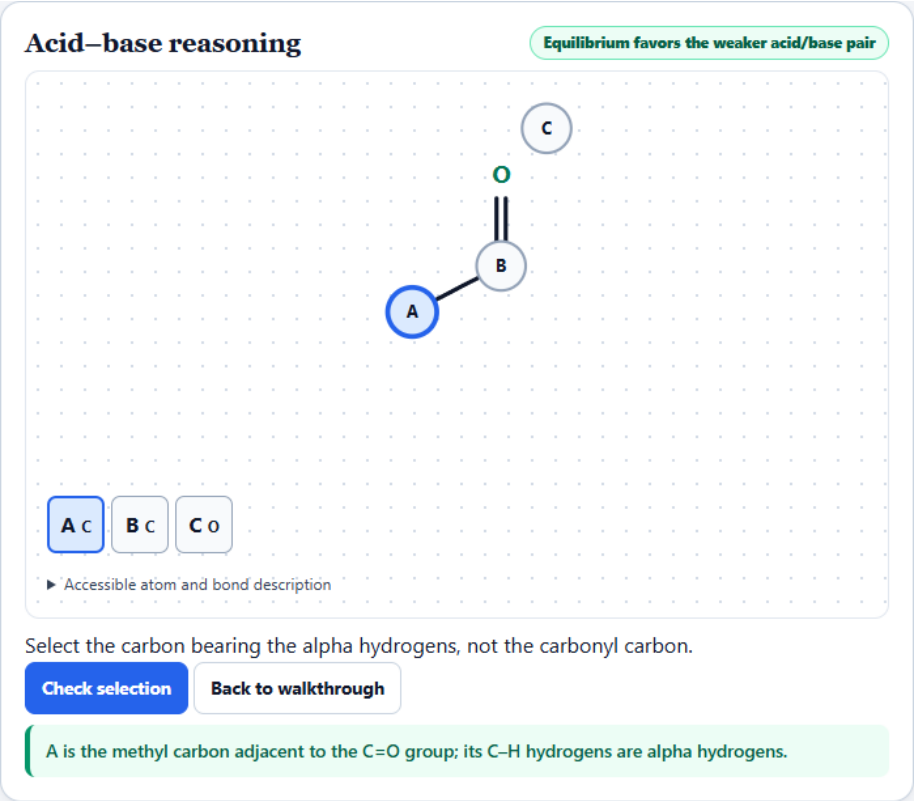
Screenshot: acids-step-3




Embedded challenge: Select the carbon bearing the alpha hydrogens, not the carbonyl carbon.

{"prompt": "Select the carbon bearing the alpha hydrogens, not the carbonyl carbon.", "answer": [0], "explanation": "A is the methyl carbon adjacent to the C=O group; its C–H hydrogens are alpha hydrogens."}

Screenshot: acids-step-3-challenge



# partners | Nucleophiles, electrophiles, and leaving groups

Electron pairs move from donor to acceptor

## partners:rule | Core Rules

A nucleophile supplies an electron pair to an electrophile. A leaving group takes the bonding pair when it departs. Stable weak bases usually make better leaving groups. Protonation can turn poor leaving group OH− into neutral water.

Schematic summary: Electron pairs move from donor to acceptor

Chemistry diagram: Follow electron donation and the leaving-group bond in this illustrative graph.

Donor — Nucleophile supplies an electron pair

Acceptor — Electrophile receives the pair

Leaving group — Takes the bonding pair on departure

## partners:cue | Memory Cues

Find electron-rich donor → electron-poor acceptor → possible leaving group. Curved arrows start at electrons, not positive charges. Compare nucleophilicity using charge, reacting-atom identity, steric access, and solvent. For a similar reacting atom, an anion is often more nucleophilic than its neutral conjugate acid. Across comparable atoms in a row, lower electronegativity often favors donation. Bulky donors are hindered in SN2. Protic solvation and polarizability can reverse simple basicity-based trends down a group; polar aprotic conditions often favor anionic SN2 nucleophiles.

Schematic summary: Choose the case that applies

Lone pair or bond — A curved arrow can start at these electrons

Positive charge alone — Not an electron source

## partners:trap | Common Mistakes / MCAT Traps

A carbonyl oxygen is electron rich while its carbon is electron poor. A positive charge is not required for electrophilicity. For comparable neutral acyl derivatives in nucleophilic acyl substitution, anhydrides are generally more reactive than esters, and esters more reactive than amides. A free carboxylic acid can instead protonate a basic nucleophile; protonation, activation and the leaving group prevent a universal acid/ester ranking.

Schematic summary: Shortcut versus a better check

Avoid — Requiring a formal positive charge for electrophilicity

Check — Carbonyl carbon is electron poor without a formal positive charge

## partners:app | MCAT Applications

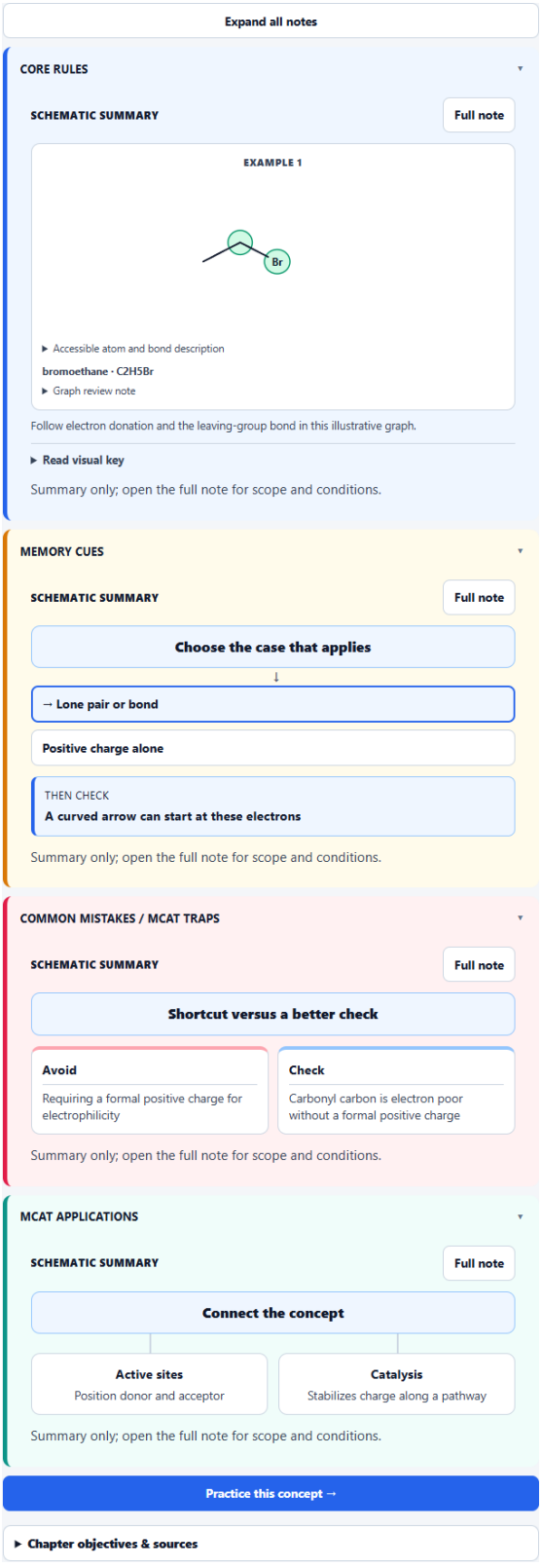
Enzyme active sites position nucleophiles and electrophiles while stabilizing charge; the same donor–acceptor logic applies.

Schematic summary: Connect the concept

Active sites — Position donor and acceptor

Catalysis — Stabilizes charge along a pathway

Screenshot: partners-visuals



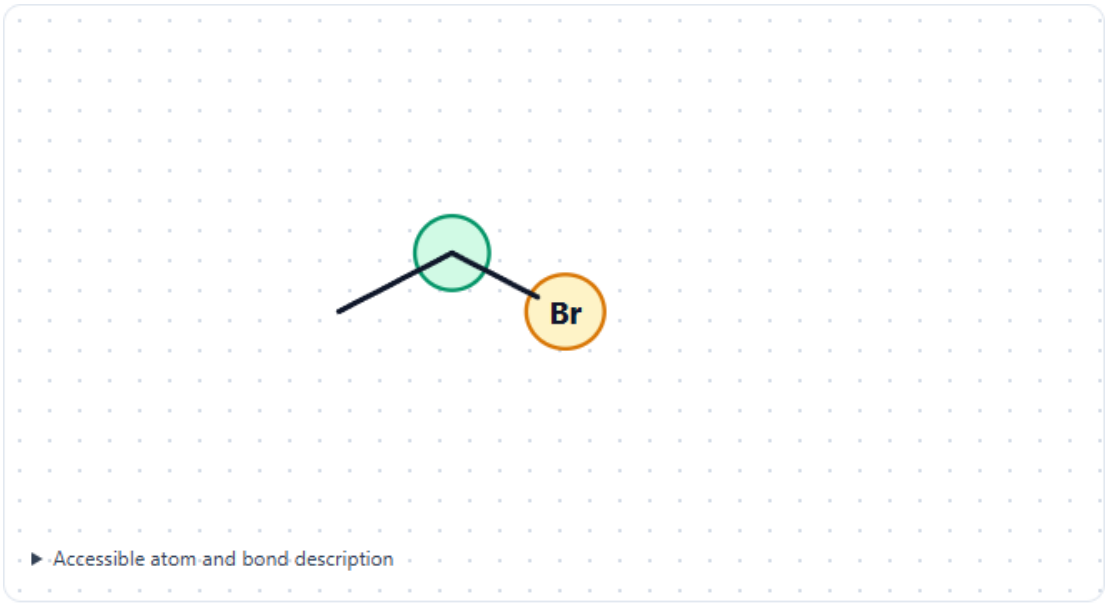

## partners:step-1 | 1. Polarize the bond

Bromine withdraws electron density from the attached carbon. The C bonded to Br is the electrophilic site for substitution.

Displayed molecule ID: bromoethane

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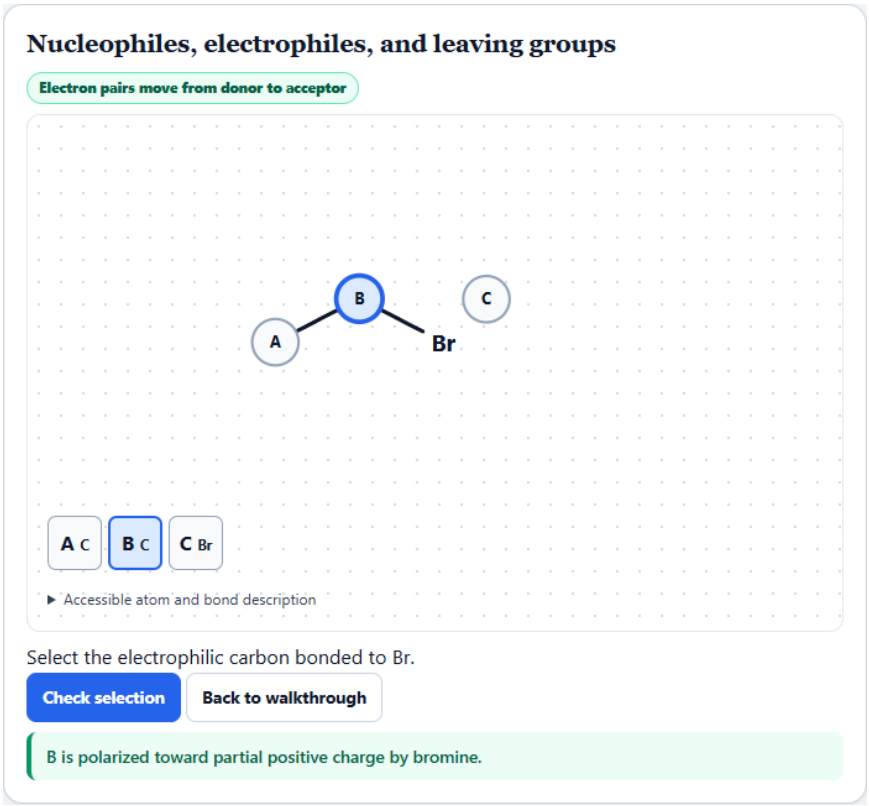
Screenshot: partners-step-1




Embedded challenge: Select the electrophilic carbon bonded to Br.

{"prompt": "Select the electrophilic carbon bonded to Br.", "answer": [1], "explanation": "B is polarized toward partial positive charge by bromine."}

Screenshot: partners-step-1-challenge



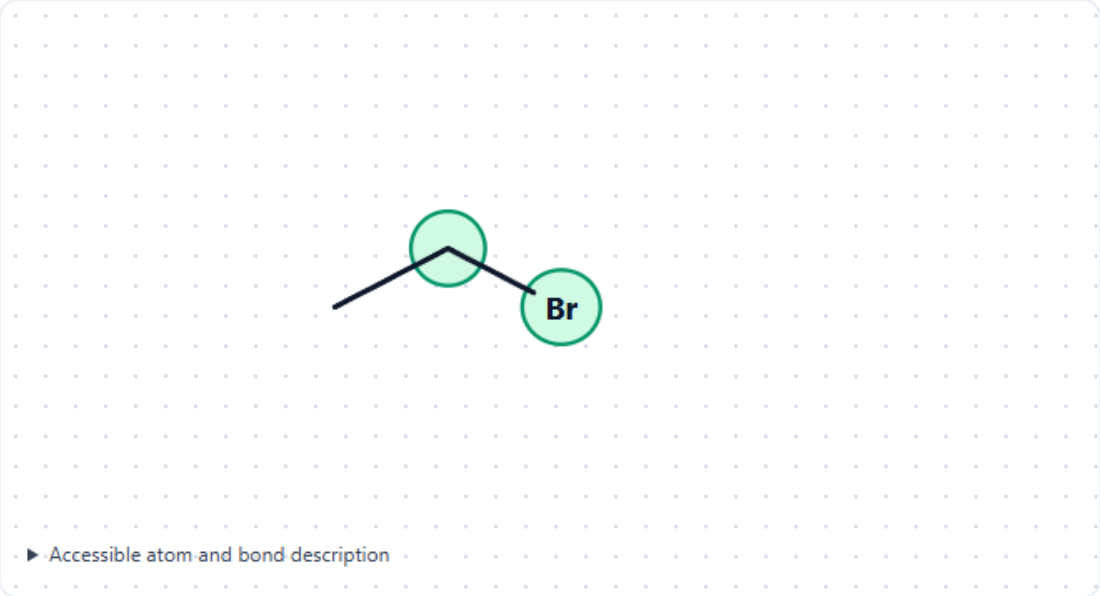

## partners:step-2 | 2. Follow bond cleavage

In substitution, a nucleophile forms a bond to the electrophilic carbon while the C–Br bond ultimately breaks with its electron pair going to Br. Conditions determine whether this is concerted or stepwise.

Displayed molecule ID: bromoethane

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: partners-step-2



# substitution | SN1 vs. SN2

Mechanism depends on substrate and conditions

## substitution:rule | Core Rules

SN2 is concerted backside attack: rate = k[substrate][nucleophile], with geometric inversion. Methyl and unhindered primary centers are favorable. SN1 ionizes to a carbocation: rate = k[substrate]; tertiary, allylic, and benzylic stabilization can help.

Schematic summary: Mechanism depends on substrate and conditions

Chemistry diagram: Compare steric access in tertiary and primary substrates; mechanism also depends on conditions.

Substrate — Steric access and carbocation stabilization

Conditions — Nucleophile, base and solvent

Mechanism — Concerted attack versus stepwise ionization

## substitution:cue | Memory Cues

SN2: two species in the rate law, one concerted event. SN1: slow ionization, then attack.

Schematic summary: Choose the case that applies

Ideal SN2 — Rate includes substrate and nucleophile

Ideal SN1 — Rate includes substrate; slow ionization precedes attack

## substitution:trap | Common Mistakes / MCAT Traps

SN1 can rearrange and often racemizes a stereocenter, but ion pairing can prevent exactly 50:50 products. SN2 inversion does not guarantee an R↔S label change if CIP priorities change. Strong bases and heat can make elimination compete.

Schematic summary: Shortcut versus a better check

Avoid — Assuming inversion always flips an R/S label

Check — CIP priorities; SN1 ion pairing can also prevent exact 50:50 products

## substitution:app | MCAT Applications

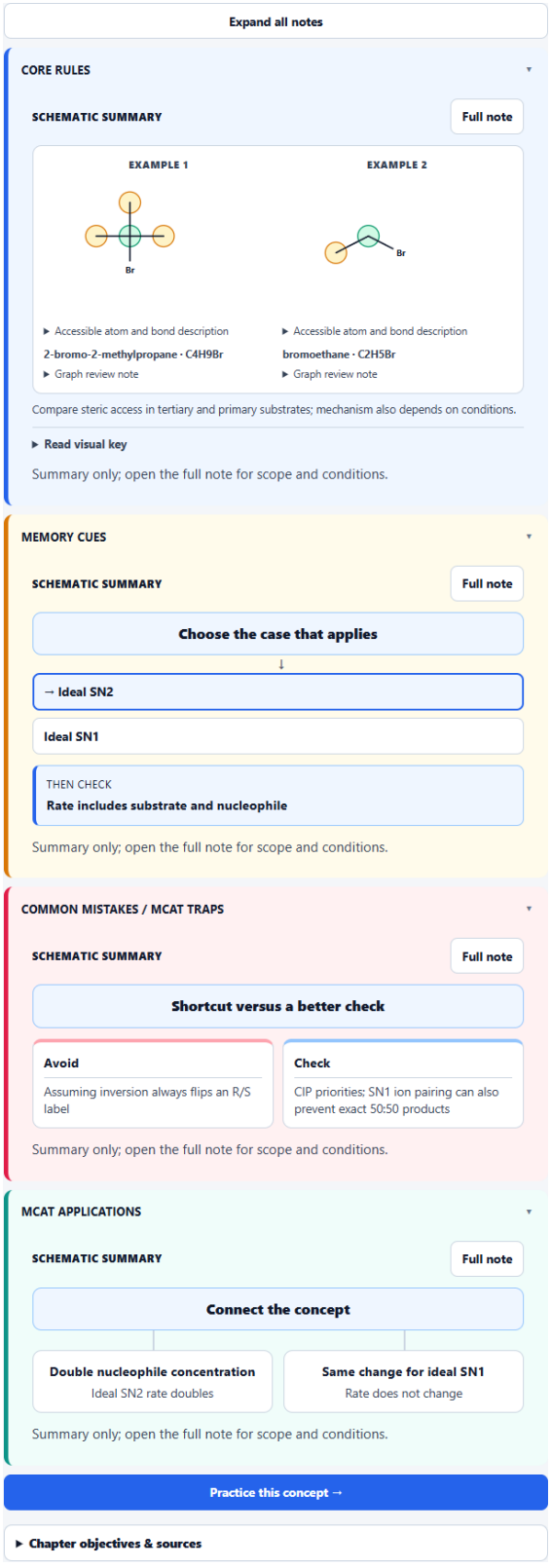
Use a rate change to distinguish mechanisms: doubling nucleophile concentration doubles an ideal SN2 rate but does not alter an ideal SN1 rate.

Schematic summary: Connect the concept

Double nucleophile concentration — Ideal SN2 rate doubles

Same change for ideal SN1 — Rate does not change

Screenshot: substitution-visuals




## substitution:step-1 | 1. Inspect crowding

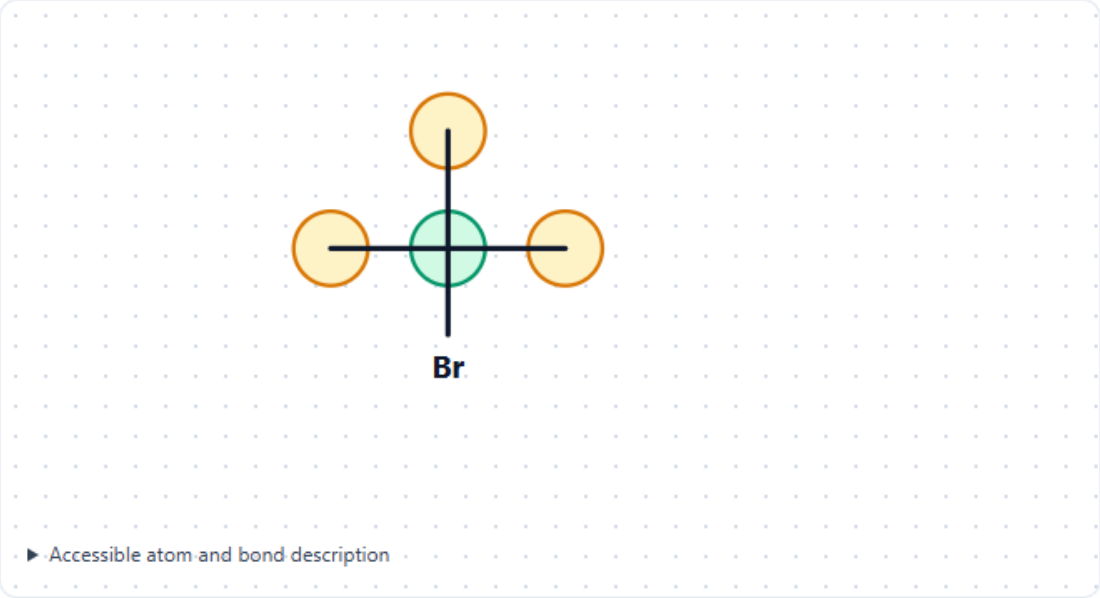
The carbon bonded to Br is tertiary: its three carbon neighbors are highlighted in orange. The cross is a 2D connectivity drawing, not a claim of 90° bond angles. Backside SN2 attack is strongly hindered; a suitable ionizing solvent can permit

formation of a tertiary carbocation.

Displayed molecule ID: tbutylbromide

{"highlights": {"prin": [1], "sub": [0, 2, 3]}, "revealName": true}

Screenshot: substitution-step-1



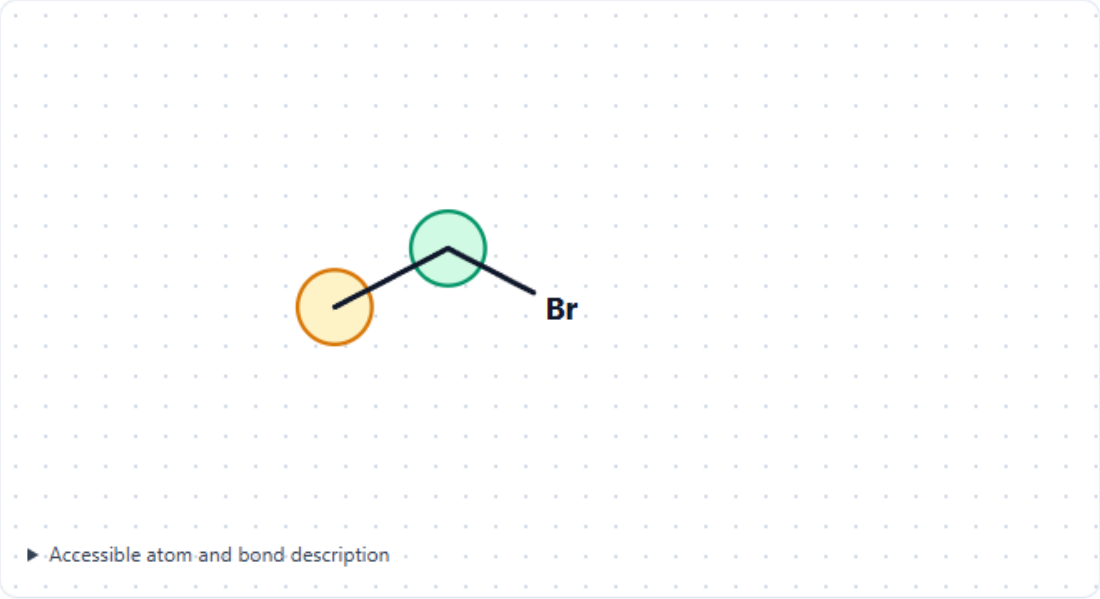

## substitution:step-2 | 2. Compare a primary site

The primary carbon has only one carbon neighbor, highlighted in orange for comparison. With a small, strong nucleophile in a suitable polar aprotic solvent, bromoethane is a better SN2 candidate. An unstabilized primary carbocation is unfavorable; strong bases or heat can make elimination compete.

Displayed molecule ID: bromoethane

{"highlights": {"prin": [1], "sub": [0]}, "revealName": true}

Screenshot: substitution-step-2



# redox | Organic oxidation and reduction

Track bonds to H and electronegative atoms

## redox:rule | Core Rules

Oxidation of carbon usually increases bonds to O/N/halogens or decreases C–H bonds. Reduction does the reverse. C–C bonds contribute zero to carbon oxidation state. PCC under anhydrous conditions can oxidize a primary alcohol to an aldehyde. Aqueous acidic $CrO_3$ or dichromate ($Na_2Cr_2O_7$/$K_2Cr_2O_7$), or suitable $KMnO_4$ conditions, can oxidize a primary alcohol further to a carboxylic acid; substrate and conditions determine the outcome.

Schematic summary: Track bonds to H and electronegative atoms

Chemistry diagram: Schematic alcohol oxidation to an aldehyde under suitable conditions; not a balanced equation.

Count carbon bonds — C–C contributes zero to carbon oxidation state

Oxidation — Usually more bonds to electronegative atoms or fewer C–H bonds

Reduction — Usually the reverse change

## redox:cue | Memory Cues

More bonds to O, less H: carbon is oxidized. Hydride addition to C=O generally reduces it.

Schematic summary: Choose the case that applies

Ordinary $NaBH_4$ — Commonly reduces aldehydes and ketones

$LiAlH_4$ with separate workup — Also acids/esters to alcohols; amides to amines

## redox:trap | Common Mistakes / MCAT Traps

$NaBH_4$ commonly reduces aldehydes and ketones but ordinarily not esters or carboxylic acids. $LiAlH_4$ reduces esters and acids to alcohols and amides to amines, using suitable nonaqueous reaction conditions followed by a separate workup. Reagent, solvent and workup matter; do not treat $LiAlH_4$ as compatible with an aqueous reaction medium.

Schematic summary: Shortcut versus a better check

Avoid — Using naming seniority as a redox ranking

Check — Actual bonds, reagent identity and conditions

## redox:app | MCAT Applications

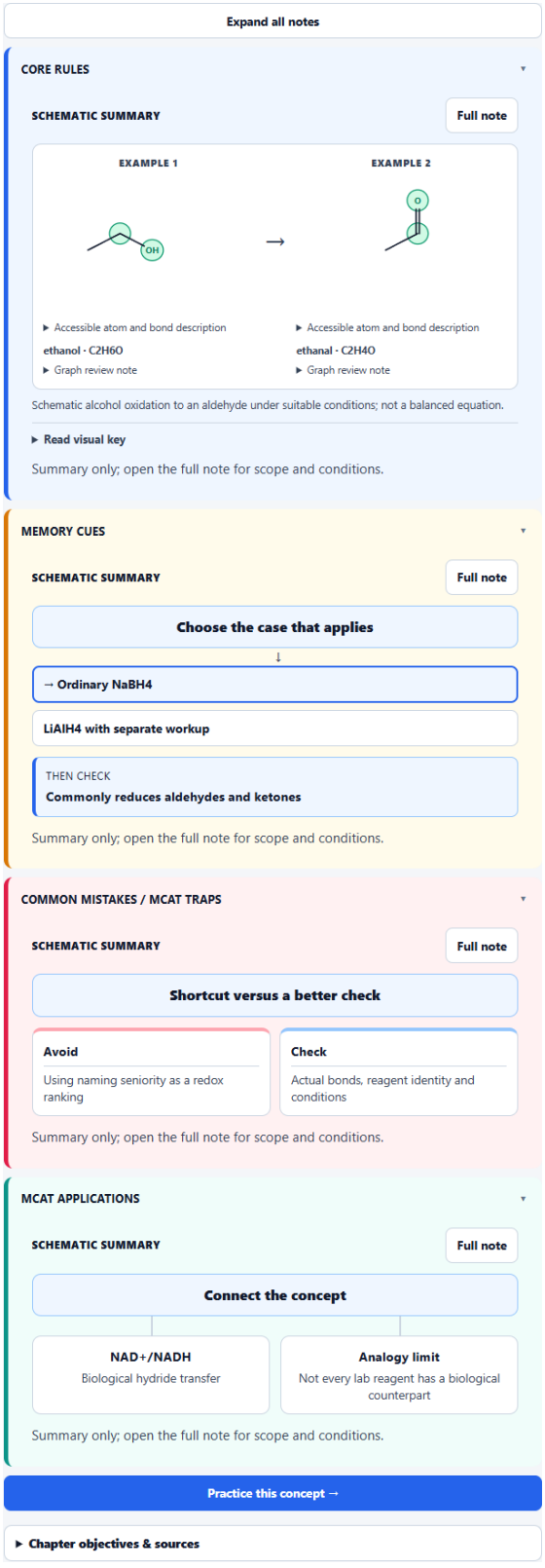
NAD+/NADH couples biological alcohol/carbonyl oxidation states through hydride transfer; not every laboratory reagent has a direct biological analogue.

Schematic summary: Connect the concept

NAD+/NADH — Biological hydride transfer

Analogy limit — Not every lab reagent has a biological counterpart

Screenshot: redox-visuals



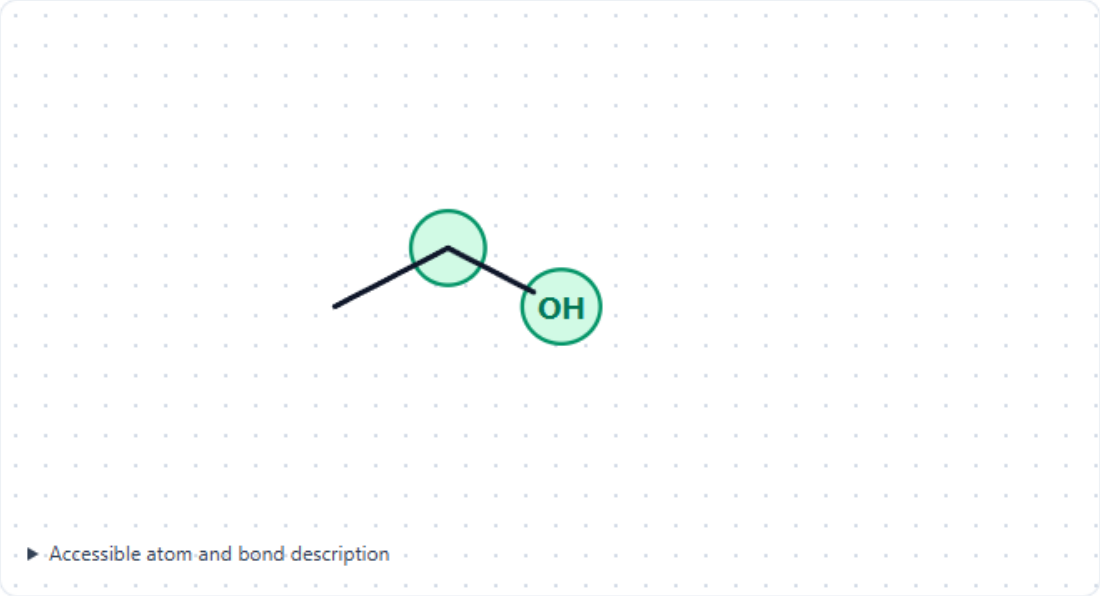

## redox:step-1 | 1. Alcohol carbon

For the CH2OH carbon, one C–O bond contributes +1 and two C–H bonds contribute −2: oxidation state −1.

Displayed molecule ID: ethanol

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: redox-step-1



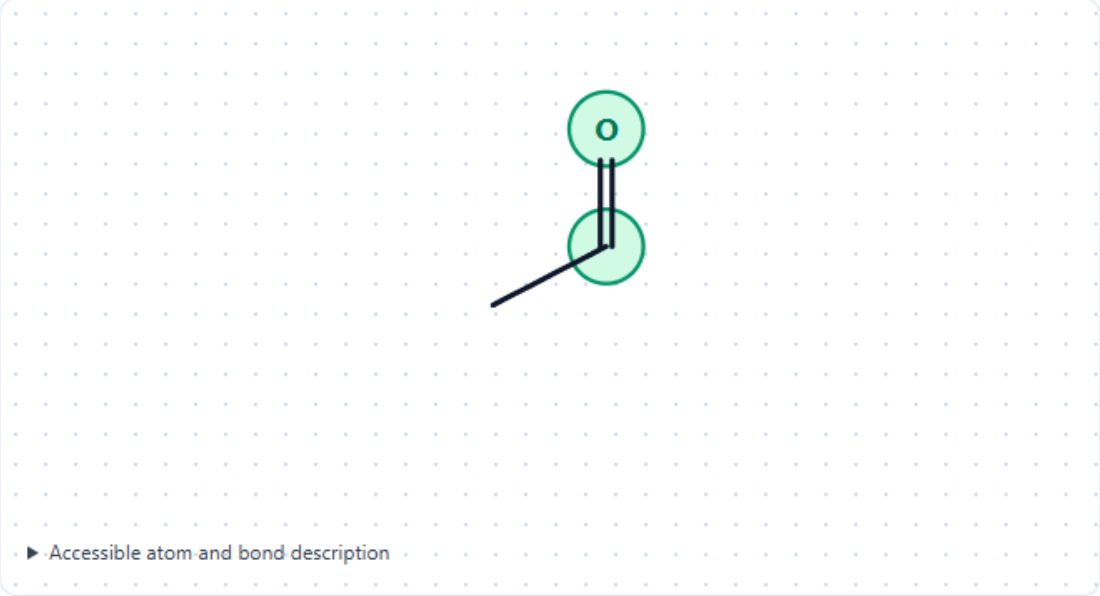

## redox:step-2 | 2. Aldehyde carbon

For CHO, C=O contributes +2, C–H contributes −1, and C–C contributes 0: oxidation state +1. Alcohol → aldehyde is a two-electron oxidation.

Displayed molecule ID: acetaldehyde

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: redox-step-2



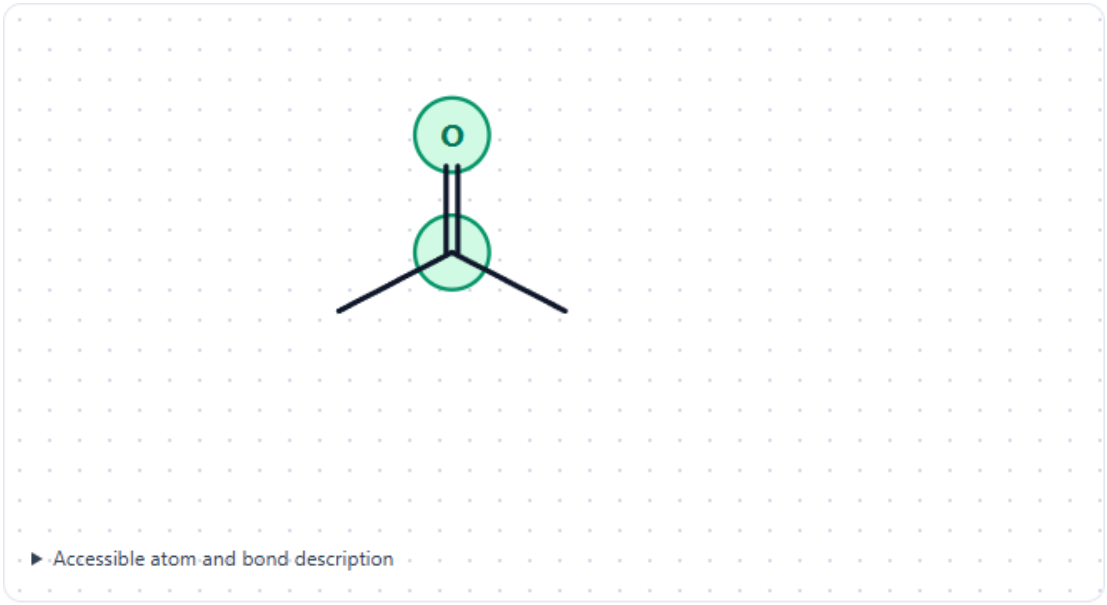

## redox:step-3 | 3. Ketone reduction

Hydride adds to the electrophilic carbonyl carbon; protonation of oxygen gives a secondary alcohol.

Displayed molecule ID: acetone

{"highlights": {"prin": [1, 3]}, "revealName": true}

Screenshot: redox-step-3



# selectivity | Chemoselectivity and a reaction workflow

Choose by reactivity, not naming seniority

## selectivity:rule | Core Rules

Inventory functional groups, identify reagent roles, locate the likely reactive site, account for solvent and acid–base steps, then track bonds and stereochemistry. Chemoselectivity means preferential reaction at one functional group.

Schematic summary: Choose by reactivity, not naming seniority

Chemistry diagram: Two functional-group foci on the same molecular graph, not a reactant/product pair.

Inventory — List functional groups and reactive sites

Reagent — Include solvent and acid–base steps

Product check — Track bonds and stereochemistry

## selectivity:cue | Memory Cues

Group inventory → reagent → first step → bond changes → product checks.

Schematic summary: Choose the case that applies

Naming seniority — Chooses a name

Reaction analysis — Uses substrate, reagent and conditions

## selectivity:trap | Common Mistakes / MCAT Traps

Do not substitute a naming-priority or carbon-oxidation-state ranking for reaction analysis. An ester is senior to a ketone in naming, yet its alkoxy oxygen donates electron density into the carbonyl by resonance. Under ordinary NaBH4 conditions, the ketone can be reduced while the ester survives. An amide is also resonance-stabilized. Acid–base reactions, activation and conditions can change which site reacts first.

Schematic summary: Shortcut versus a better check

Avoid — Senior in naming therefore more reactive

Check — Mechanism, resonance and reagent scope

## selectivity:app | MCAT Applications

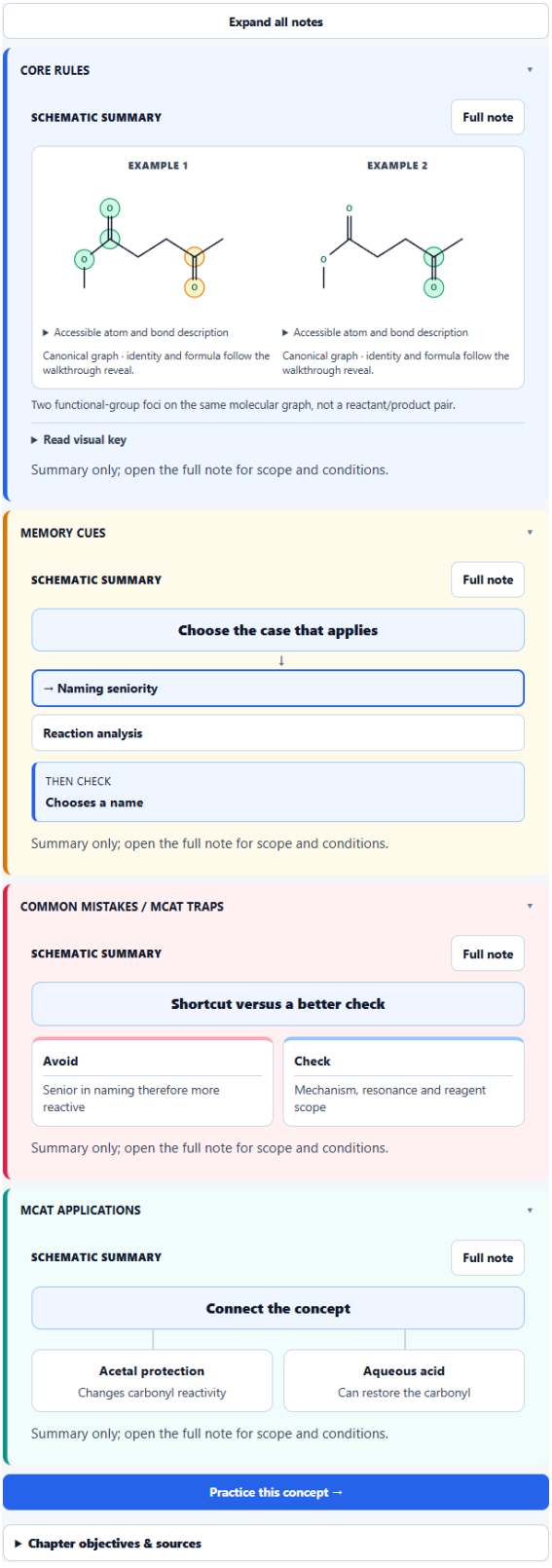
Acetals protect aldehyde/ketone carbonyls from many basic and nucleophilic conditions; aqueous acid restores the carbonyl. Protection replaces the carbonyl pi system with a different functional group, changing electronic reactivity rather than merely adding steric bulk. Carbonyl compounds may also react through acidic alpha hydrogens: deprotonation creates an enolate nucleophile instead of nucleophilic addition at the carbonyl carbon.

Schematic summary: Connect the concept

Acetal protection — Changes carbonyl reactivity

Aqueous acid — Can restore the carbonyl

Screenshot: selectivity-visuals



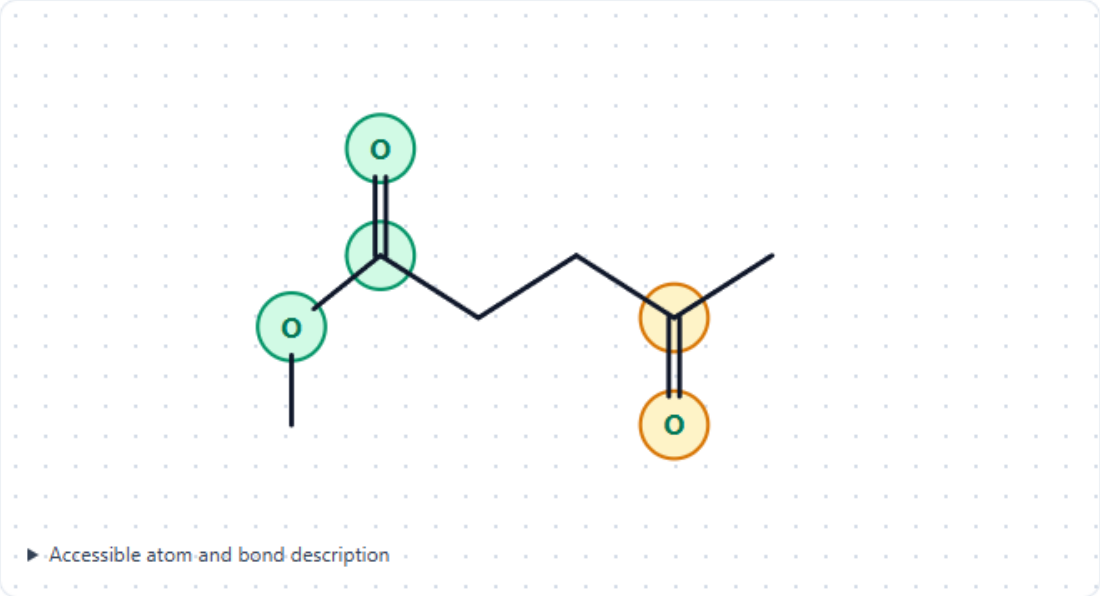

## selectivity:step-1 | 1. Inventory two carbonyls

This molecule contains an ester at the left and a ketone at the right. The ester has an oxygen substituent directly on its carbonyl carbon.

Displayed molecule ID: ketoester

{"highlights": {"prin": [0, 5, 6], "sub": [3, 8]}, "revealName": true}

Screenshot: selectivity-step-1



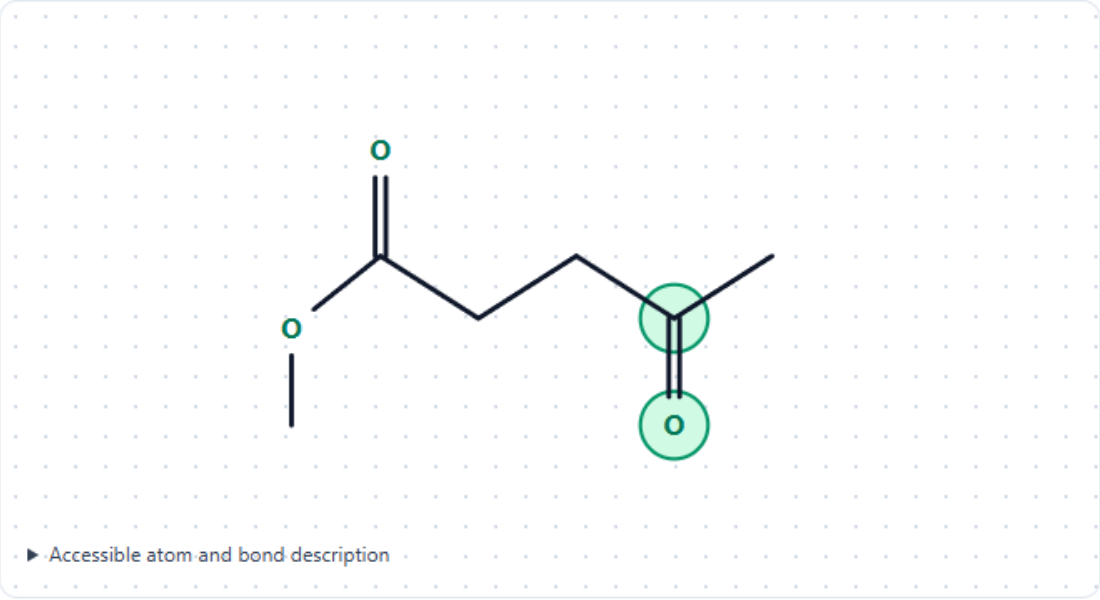

## selectivity:step-2 | 2. Predict a selective step

Ordinary NaBH4 conditions favor reduction of the ketone. Ester resonance donation lowers its electrophilicity relative to the ketone; a stronger reducing system can change the outcome.

Displayed molecule ID: ketoester

{"highlights": {"prin": [3, 8]}, "revealName": false}

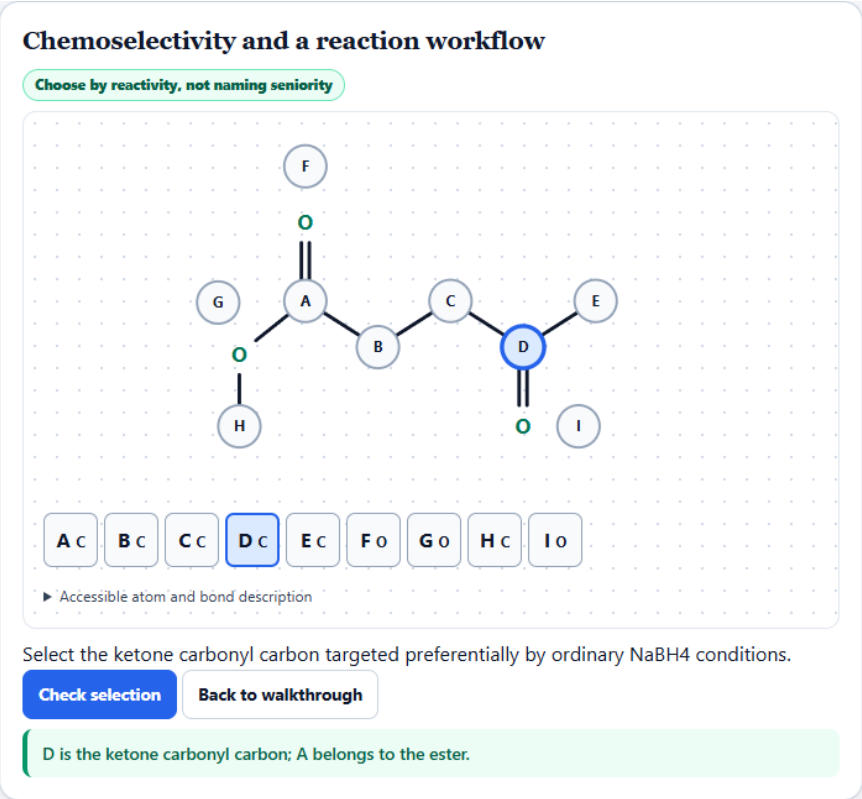
Screenshot: selectivity-step-2




Embedded challenge: Select the ketone carbonyl carbon targeted preferentially by ordinary NaBH4 conditions.

{"prompt": "Select the ketone carbonyl carbon targeted preferentially by ordinary NaBH4 conditions.", "answer": [3], "explanation": "D is the ketone carbonyl carbon; A belongs to the ester."}

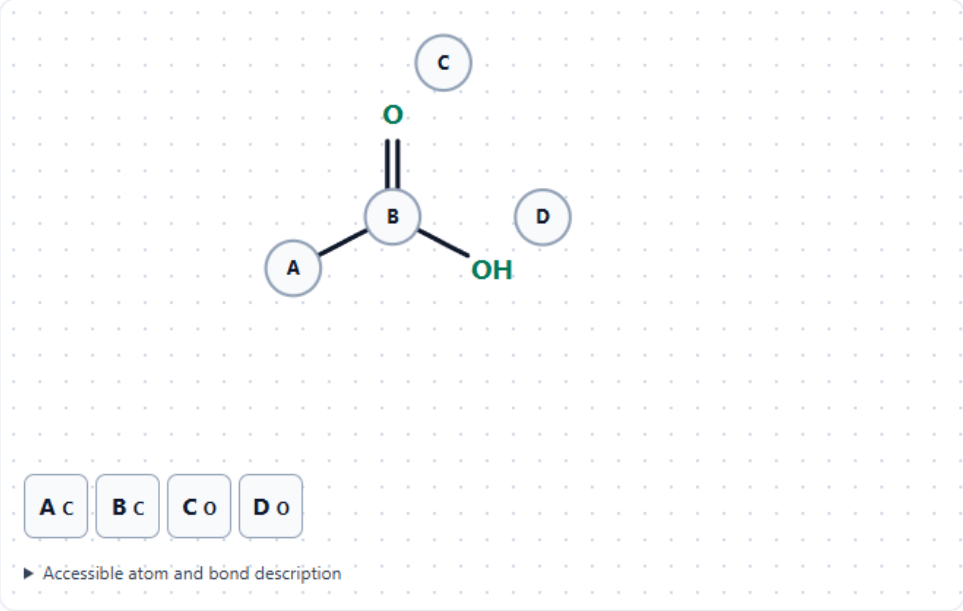
Screenshot: selectivity-step-2-challenge



# pka-equilibrium | Estimate acid–base equilibrium

The displayed carboxyl group is a structural example for Stage 1. Stages 2–3 use a separate hypothetical proton-transfer equilibrium, HA + B ⇌ A− + BH+, with all pKa values on a comparable solvent/temperature scale; they are not measured values for the displayed ethanoic acid.

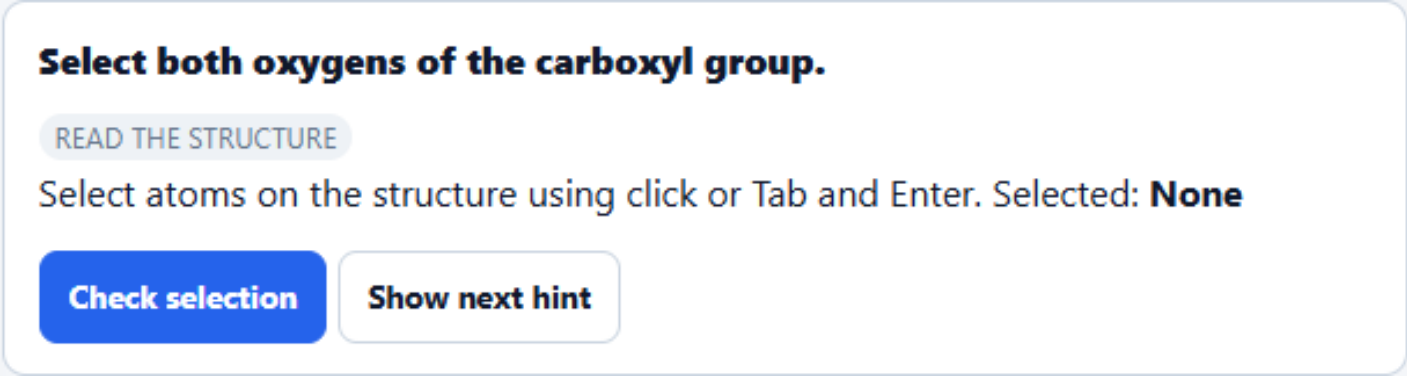
Screenshot: pka-equilibrium-stage-1-molecule




## pka-equilibrium:stage-1 | Select both oxygens of the carboxyl group.

Answer type: atoms

Screenshot: pka-equilibrium-stage-1-question



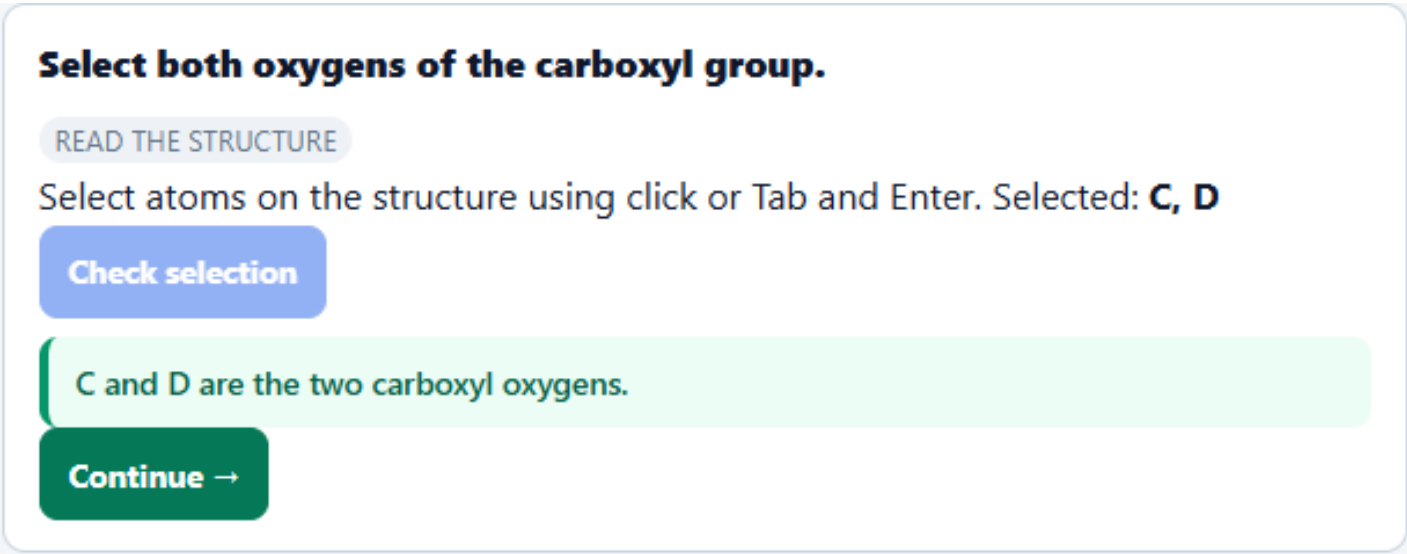

{"authorAnswer": [2, 3], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [2, 3]}}

Hint 1: Locate C=O and O–H on the same carbon.

Hint 2: Select the two O labels.

Explanation: C and D are the two carboxyl oxygens.

Screenshot: pka-equilibrium-stage-1-feedback




### pka-equilibrium:stage-2 | For HA + B ⇌ A− + BH+, if pKa(HA)=5 and pKa(BH+)=10, equilibrium favors…

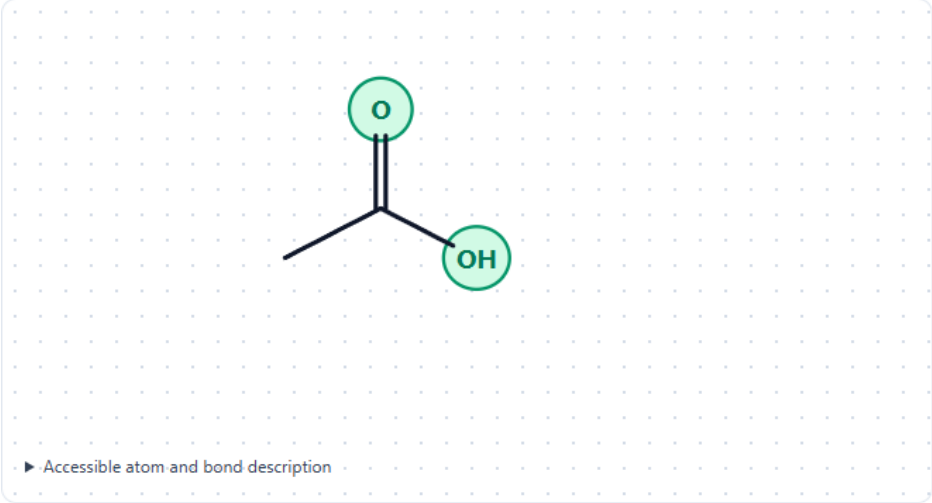
Answer type: choice

Option 0 (A): Reactants

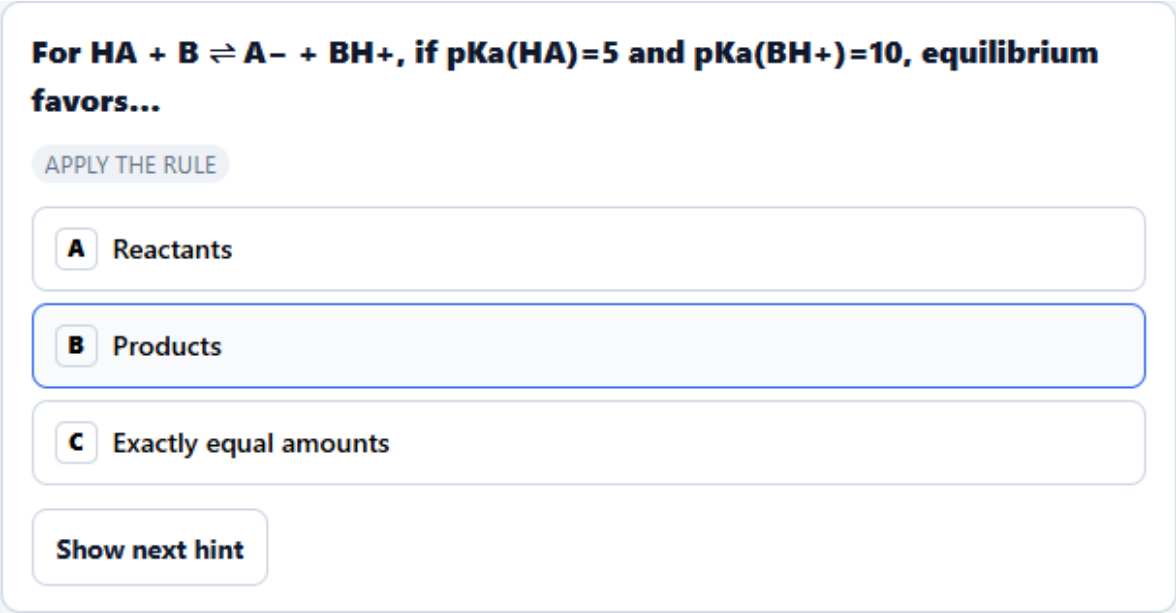
Option 1 (B): Products

Option 2 (C): Exactly equal amounts

Screenshot: pka-equilibrium-stage-2-molecule



Screenshot: pka-equilibrium-stage-2-question



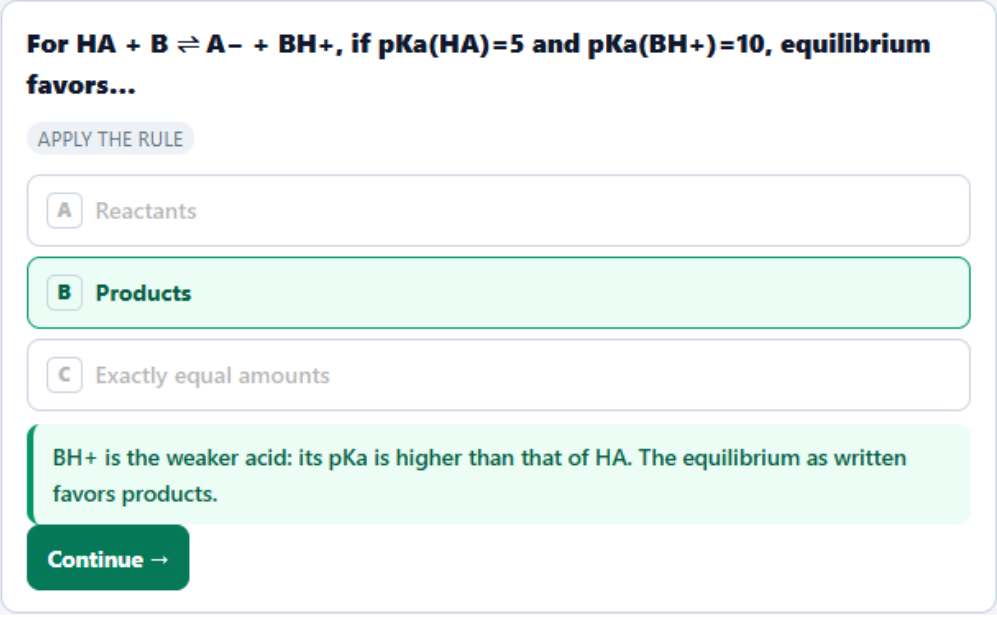

{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Compare the two acids.

Hint 2: Equilibrium favors the side with the higher-pKa acid.

Explanation: BH+ is the weaker acid: its pKa is higher than that of HA. The equilibrium as written favors products.

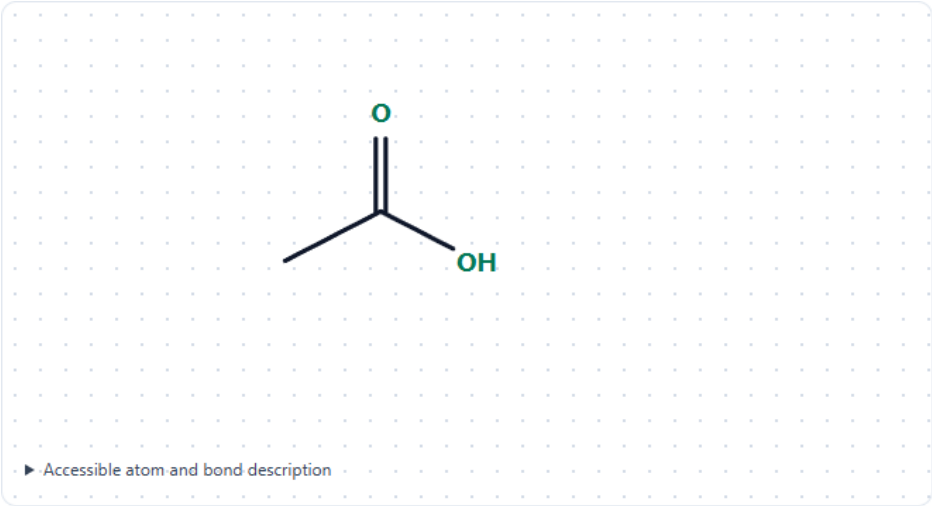
Screenshot: pka-equilibrium-stage-2-feedback



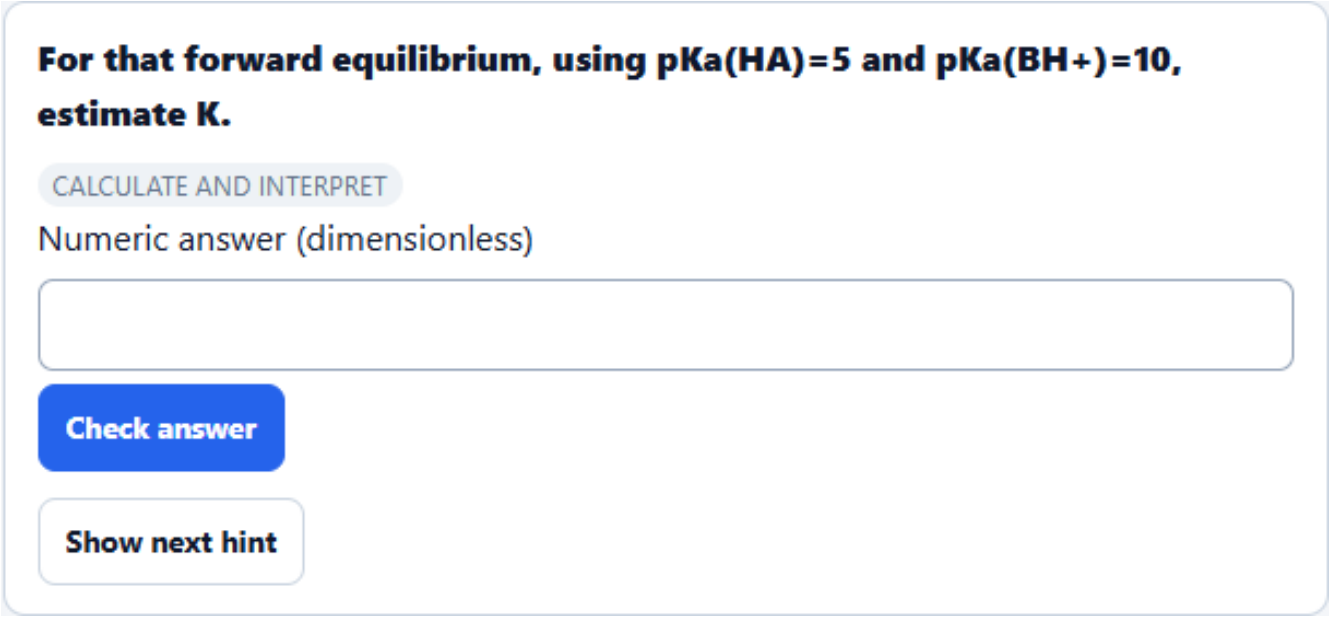

## pka-equilibrium:stage-3 | For that forward equilibrium, using pKa(HA)=5 and pKa(BH+)=10, estimate K.

Answer type: number | unit: dimensionless

Screenshot: pka-equilibrium-stage-3-molecule




Screenshot: pka-equilibrium-stage-3-question



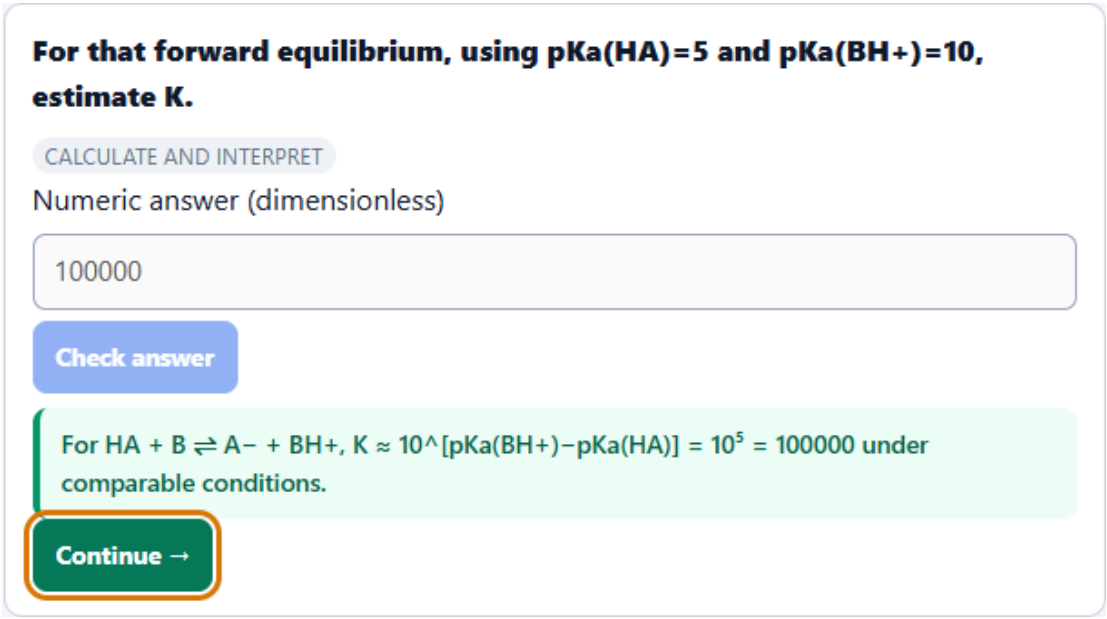

{"authorAnswer": 100000, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Subtract reactant-acid pKa from product-acid pKa.

Hint 2: Raise ten to that difference.

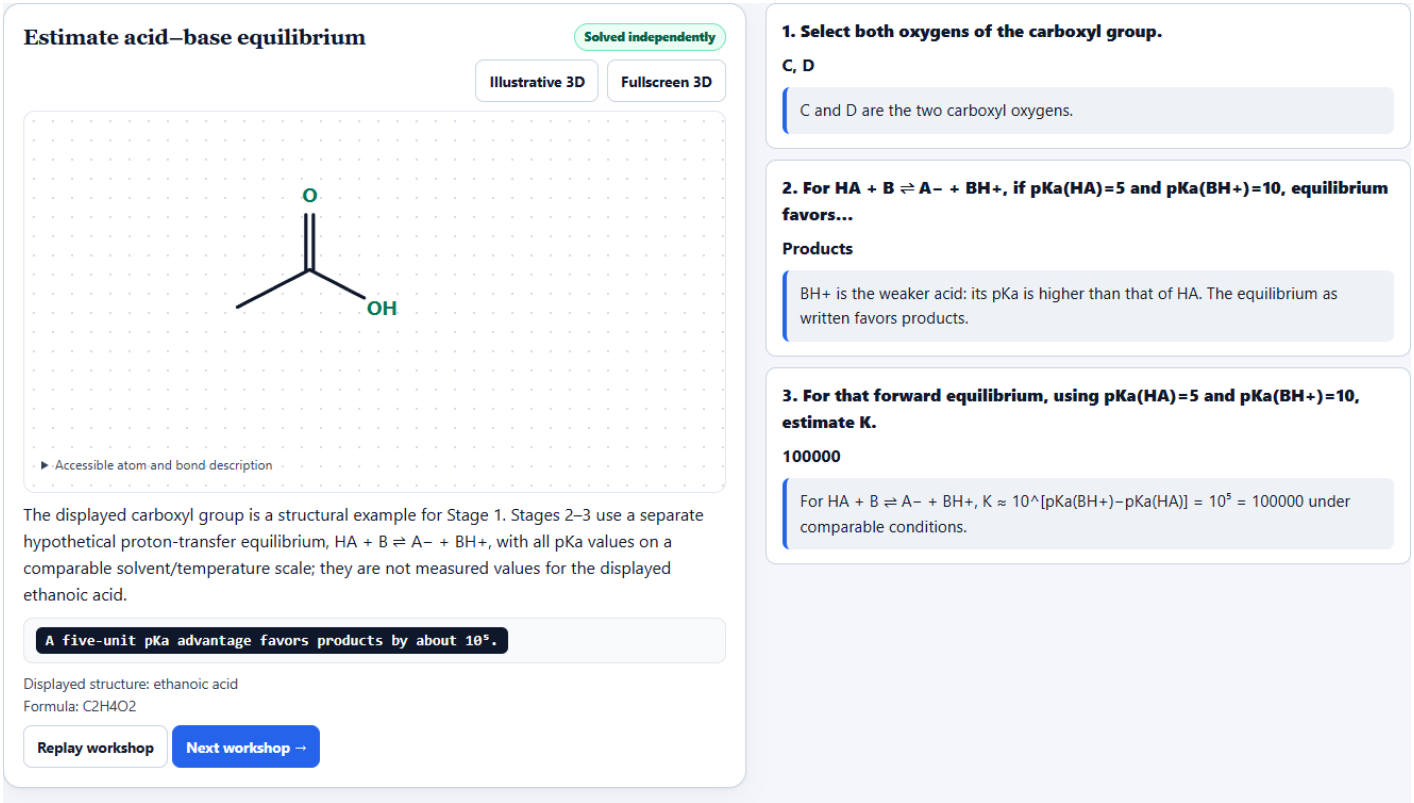
Explanation: For HA + B ⇌ A− + BH+, K ≈ 10^[pKa(BH+)−pKa(HA)] = $10^5$ = 100000 under comparable conditions.

Screenshot: pka-equilibrium-stage-3-feedback



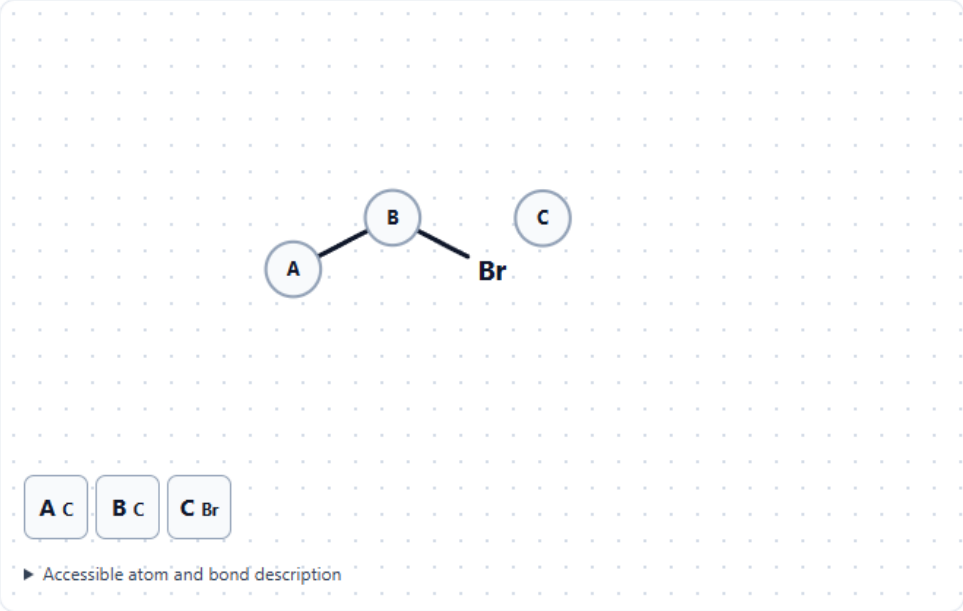

Completion summary: A five-unit pKa advantage favors products by about $10^5$.

Screenshot: pka-equilibrium-complete



# identify-partners | Find the substitution partners

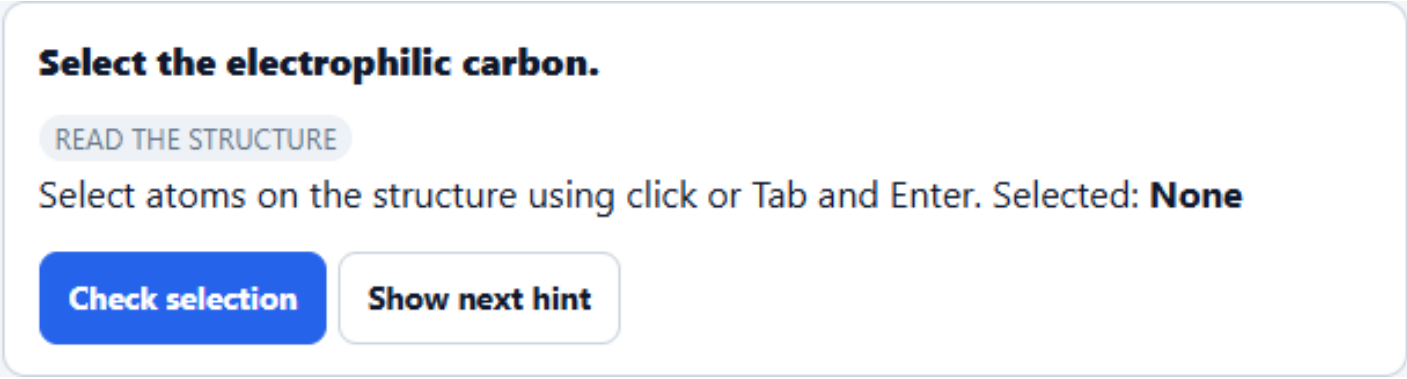
Screenshot: identify-partners-stage-1-molecule




## identify-partners:stage-1 | Select the electrophilic carbon.

Answer type: atoms

Screenshot: identify-partners-stage-1-question



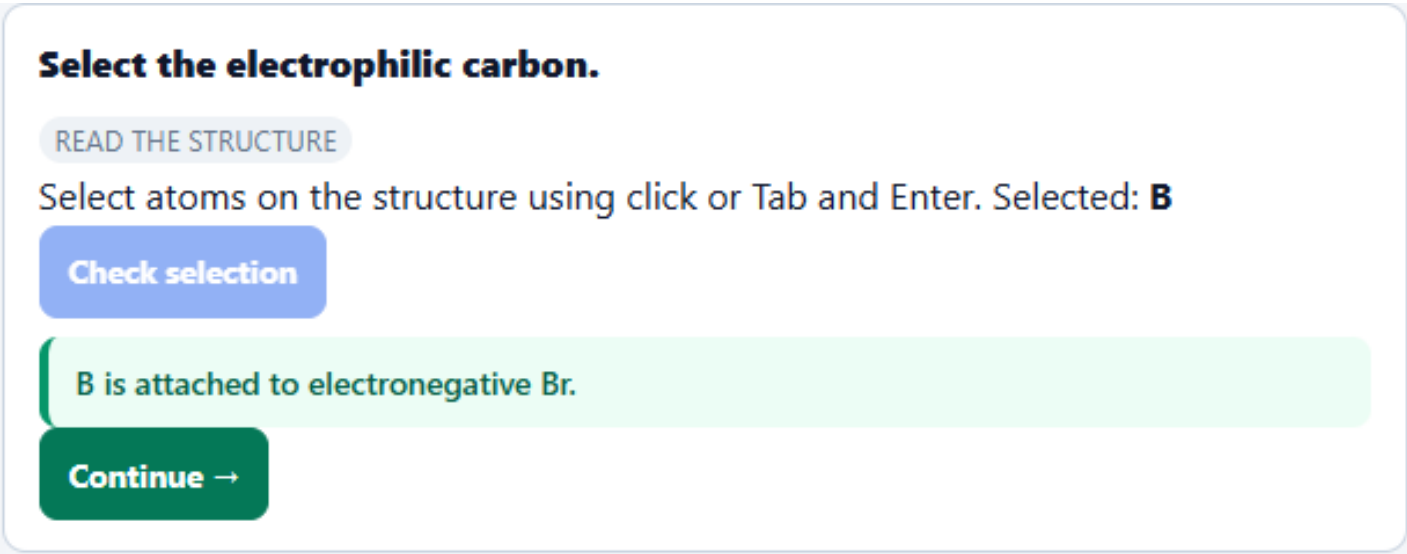

{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1]}}

Hint 1: Look for the carbon bonded to the leaving group.

Hint 2: The polarized C–Br bond identifies it.

Explanation: B is attached to electronegative Br.

Screenshot: identify-partners-stage-1-feedback




### identify-partners:stage-2 | Among these choices (including a reagent not drawn), which species supplies a lone pair as a nucleophile?

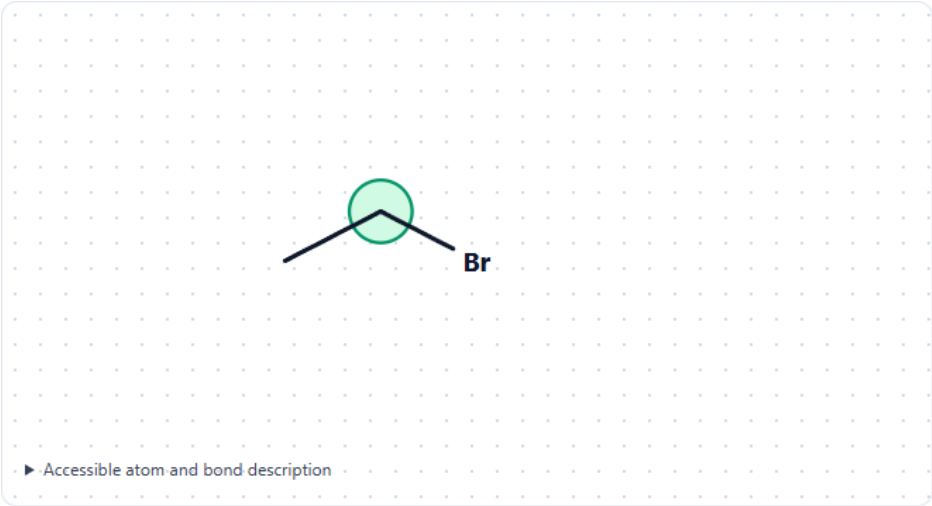
Answer type: choice

Option 0 (A): Hydroxide ion

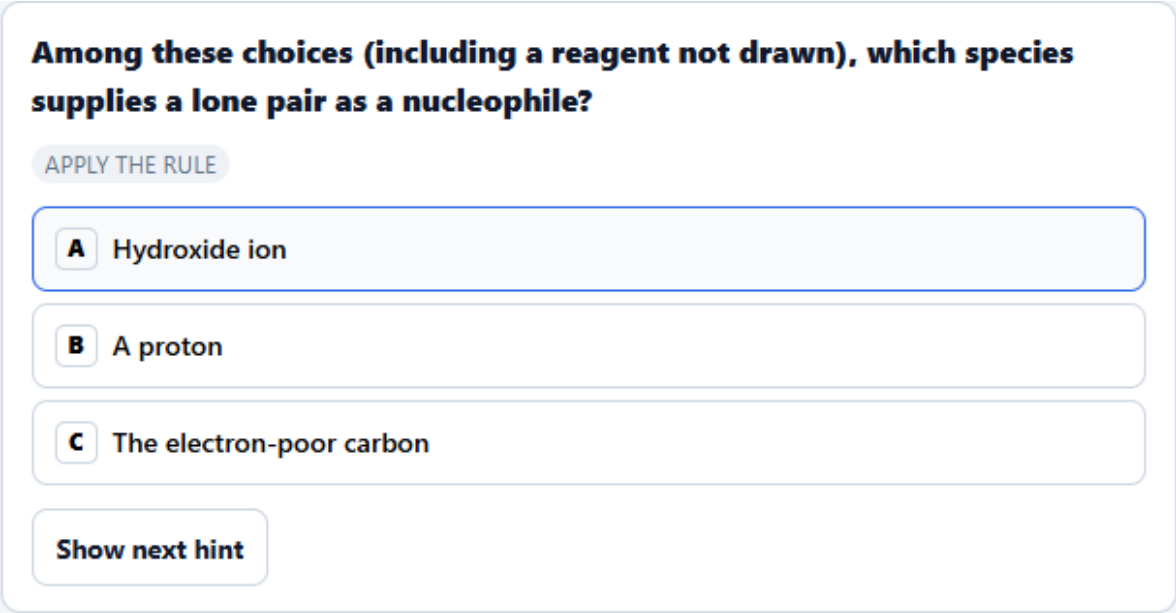
Option 1 (B): A proton

Option 2 (C): The electron-poor carbon

Screenshot: identify-partners-stage-2-molecule



Screenshot: identify-partners-stage-2-question



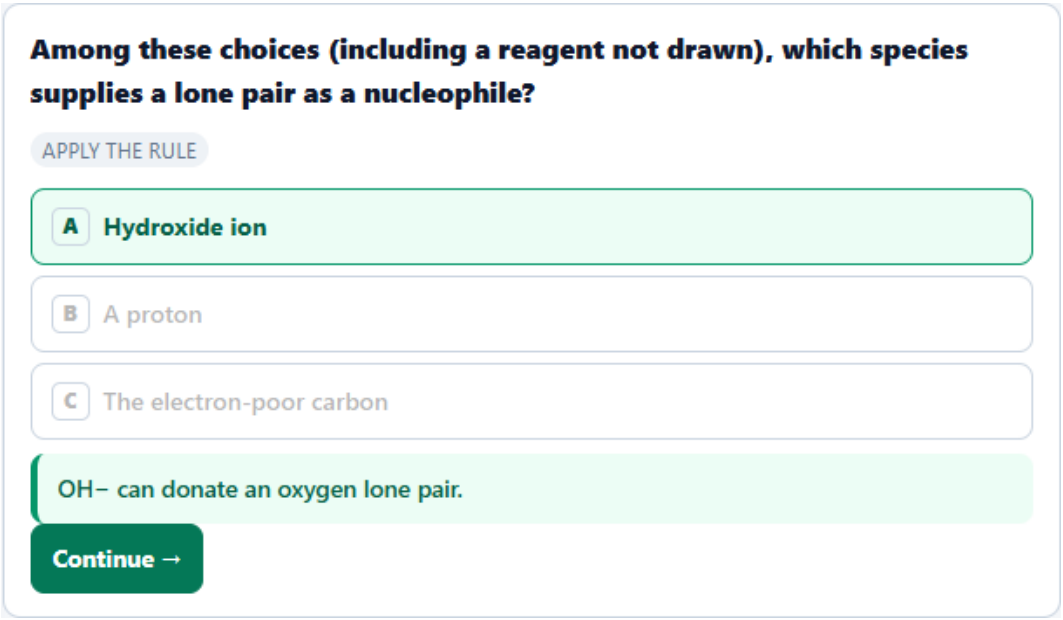

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Nucleophiles are electron-pair donors.

Hint 2: A negative charge often indicates available electrons.

Explanation: OH− can donate an oxygen lone pair.

Screenshot: identify-partners-stage-2-feedback




## identify-partners:stage-3 | When C–Br breaks heterolytically, the pair goes to…

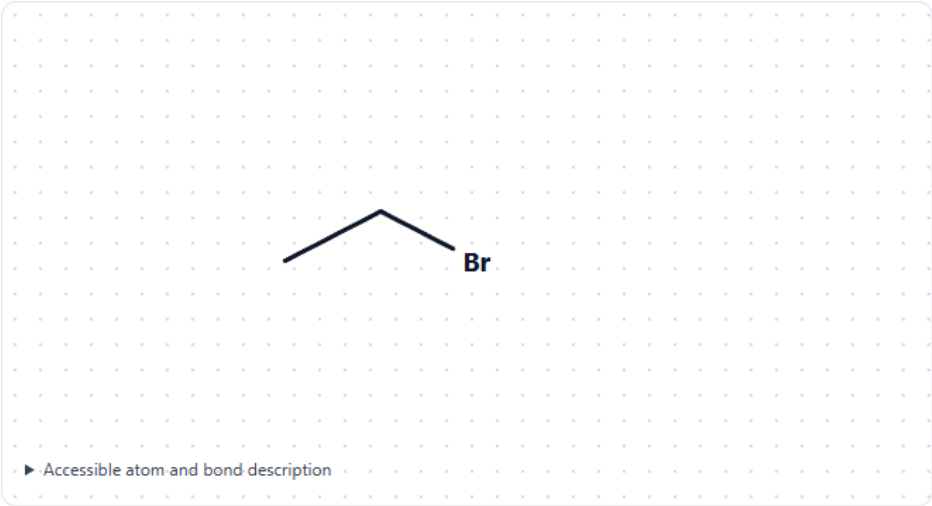
Answer type: choice

Option 0 (A): Carbon

Option 1 (B): Bromine

Option 2 (C): Neither atom

Screenshot: identify-partners-stage-3-molecule




Screenshot: identify-partners-stage-3-question



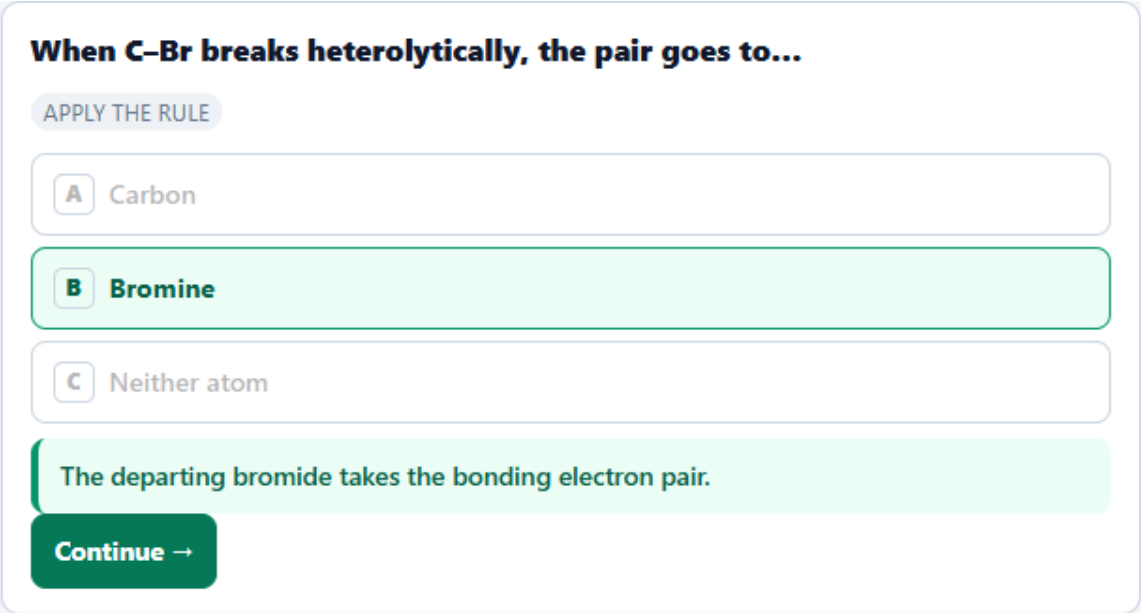

{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: A leaving group retains the bond electrons.

Hint 2: Bromide is a relatively stable anion.

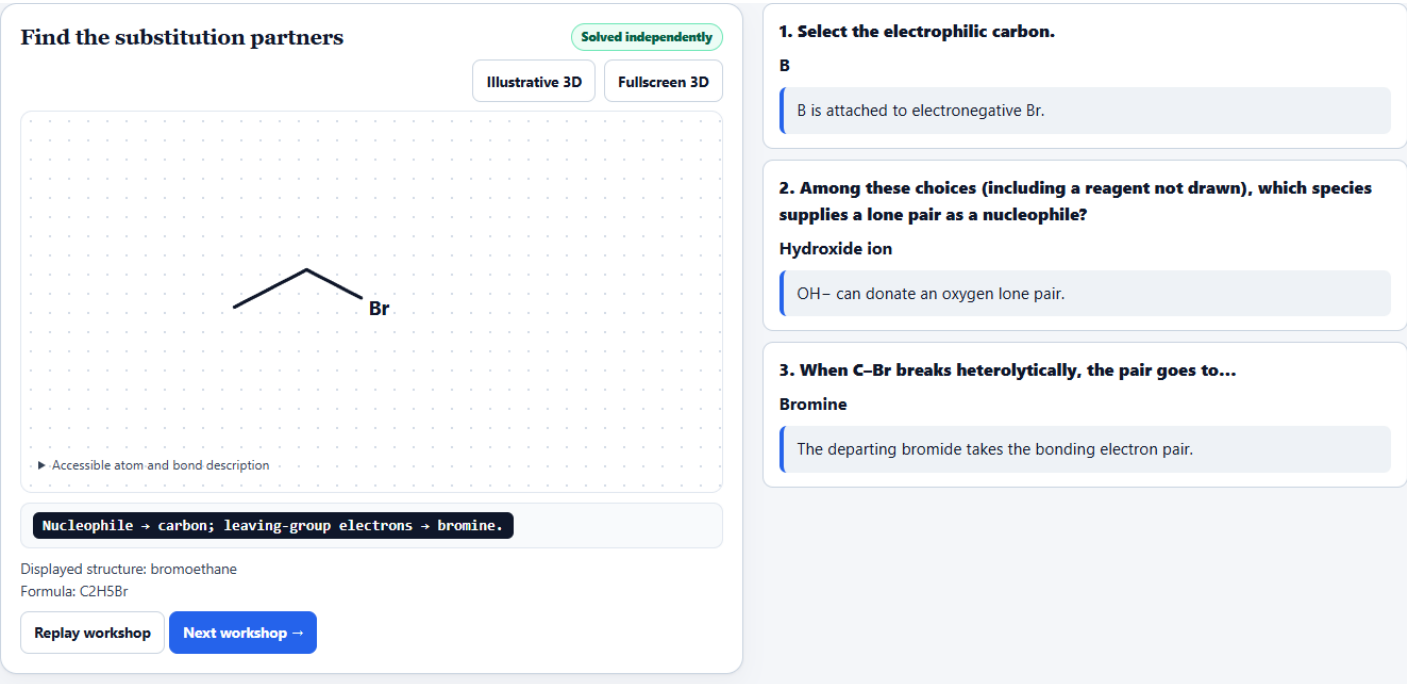
Explanation: The departing bromide takes the bonding electron pair.

Screenshot: identify-partners-stage-3-feedback



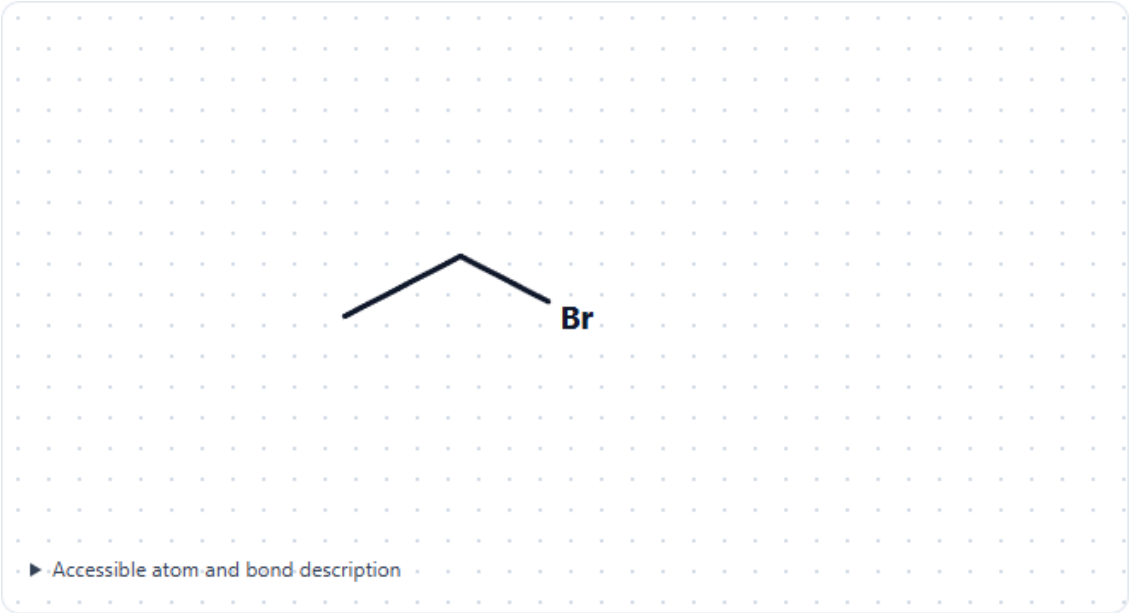

Completion summary: Nucleophile → carbon; leaving-group electrons → bromine.

Screenshot: identify-partners-complete



# sn2-rate | Mechanism and kinetics

Screenshot: sn2-rate-stage-1-molecule



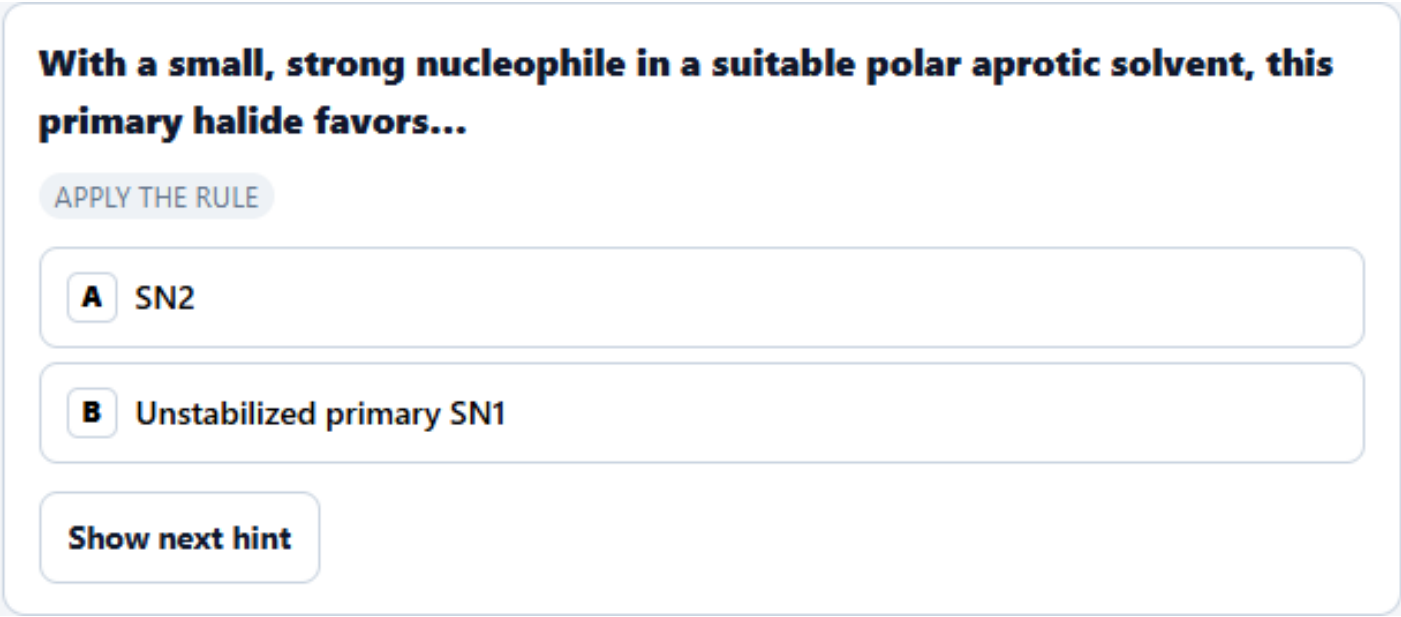

## sn2-rate:stage-1 | With a small, strong nucleophile in a suitable polar aprotic solvent, this primary halide favors...

Answer type: choice

Option 0 (A): SN2

Option 1 (B): Unstabilized primary SN1

Screenshot: sn2-rate-stage-1-question



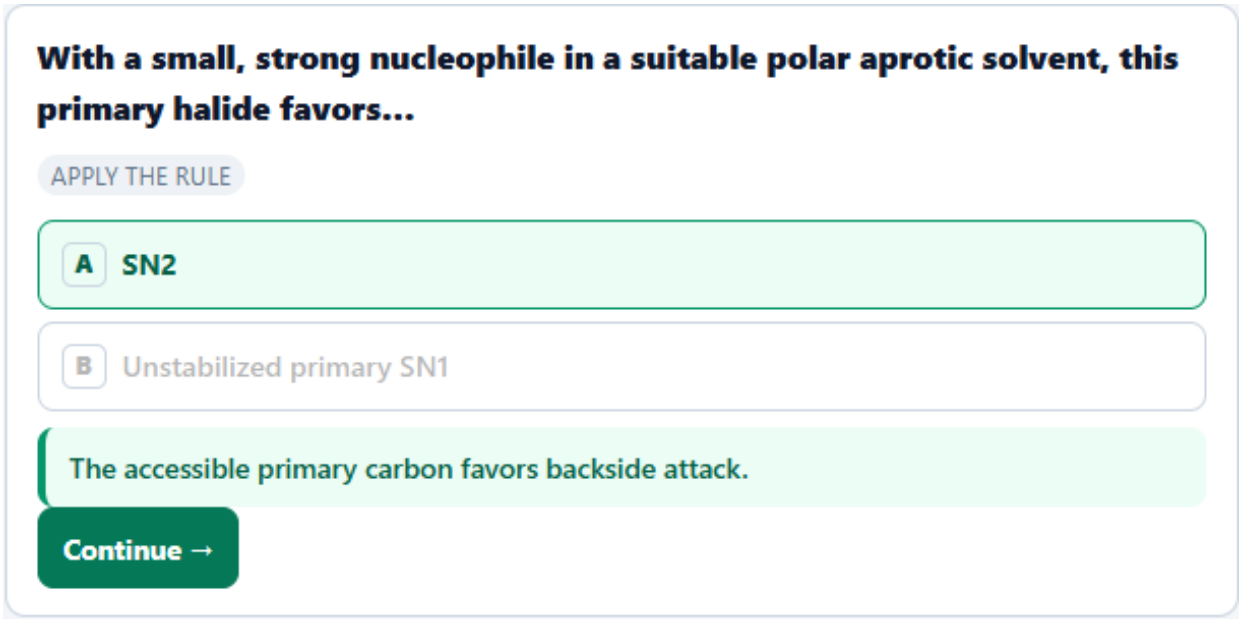

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Assess steric access and carbocation stability.

Hint 2: An unstabilized primary carbocation is poor.

Explanation: The accessible primary carbon favors backside attack.

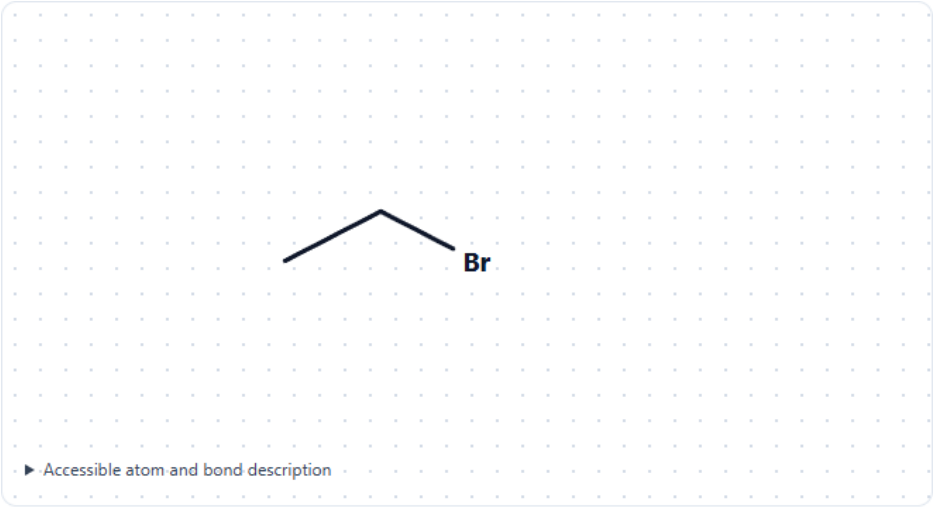
Screenshot: sn2-rate-stage-1-feedback



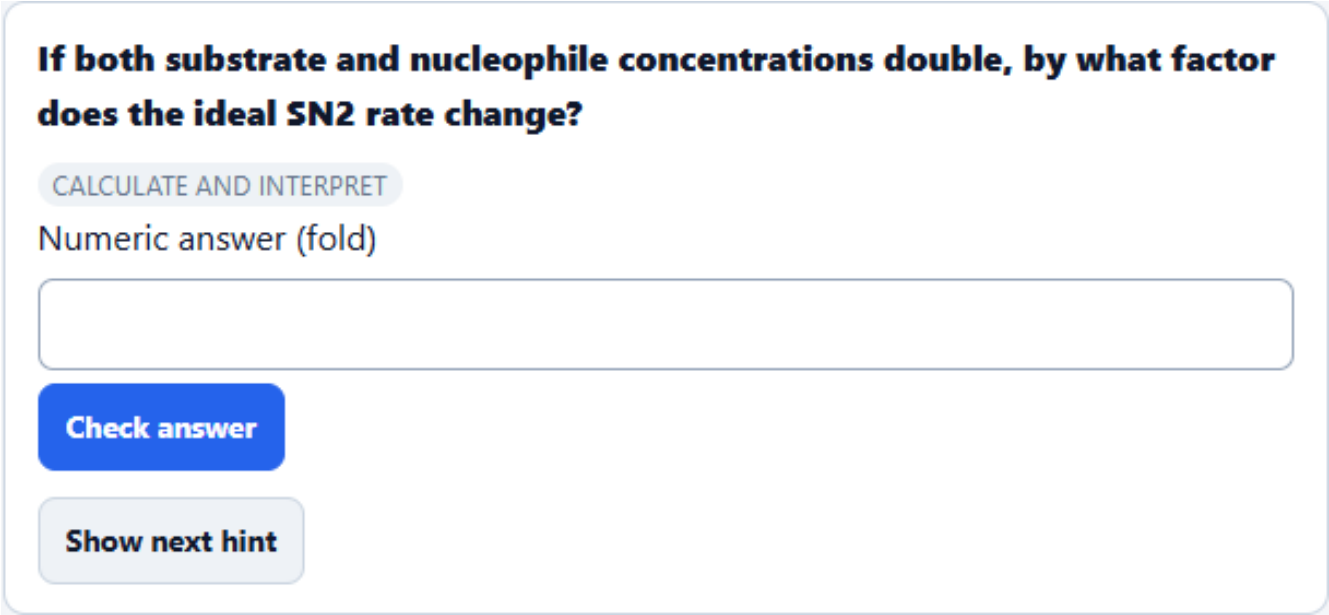

## sn2-rate:stage-2 | If both substrate and nucleophile concentrations double, by what factor does the ideal SN2 rate change?

Answer type: number | unit: fold

Screenshot: sn2-rate-stage-2-molecule



Screenshot: sn2-rate-stage-2-question



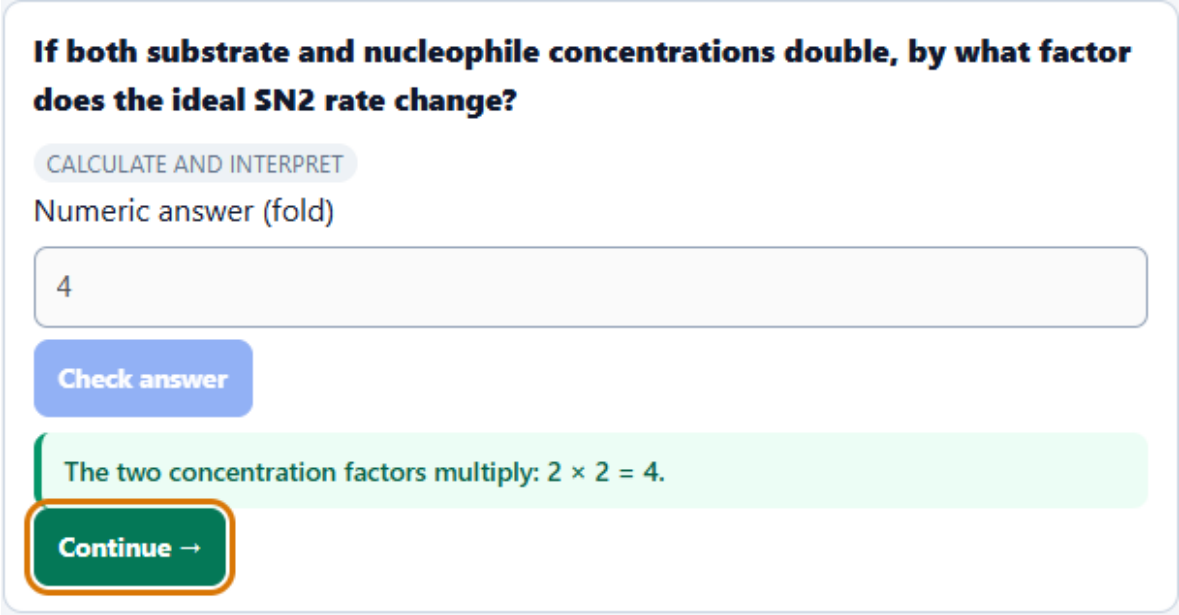

{"authorAnswer": 4, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use rate = k[substrate][nucleophile].

Hint 2: Each concentration contributes one factor of two.

Explanation: The two concentration factors multiply: 2 × 2 = 4.

Screenshot: sn2-rate-stage-2-feedback



## sn2-rate:stage-3 | Apply the general rule to a separate chiral SN2 center (the displayed bromoethane is achiral): its geometry undergoes…

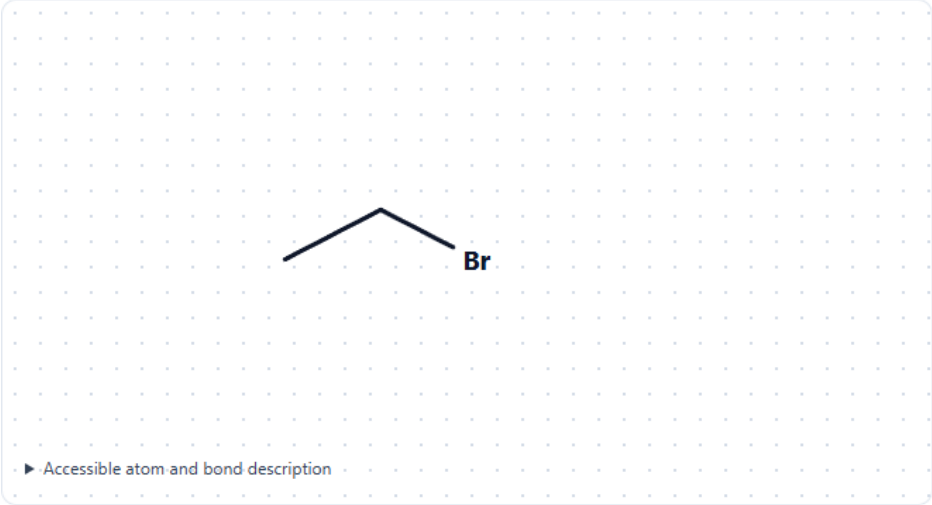
Answer type: choice

Option 0 (A): Inversion

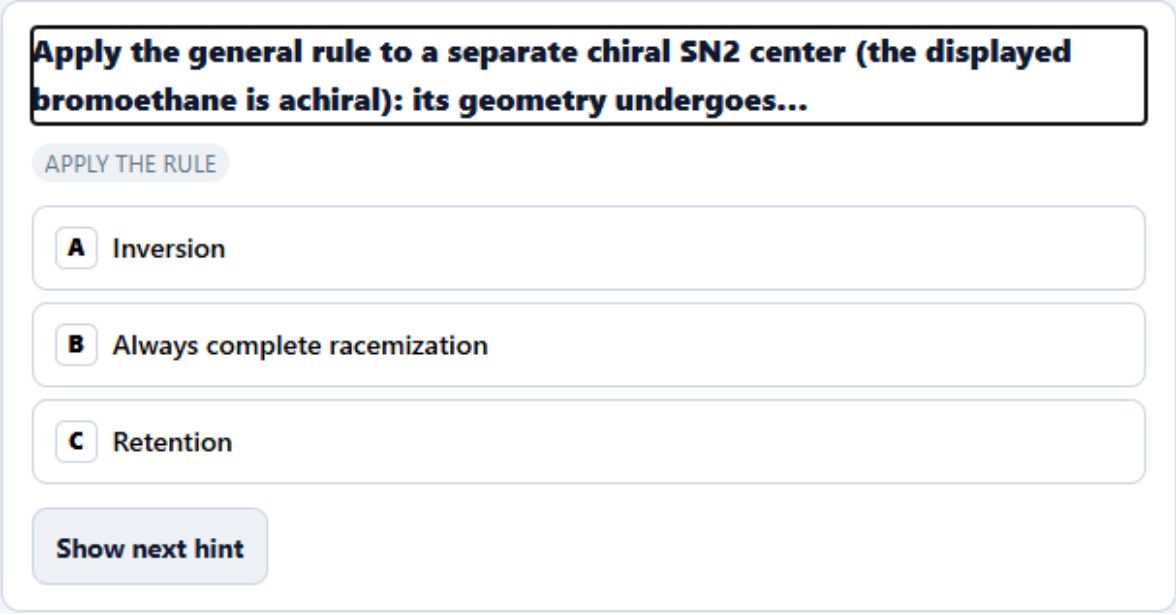
Option 1 (B): Always complete racemization

Option 2 (C): Retention

Screenshot: sn2-rate-stage-3-molecule




Screenshot: sn2-rate-stage-3-question



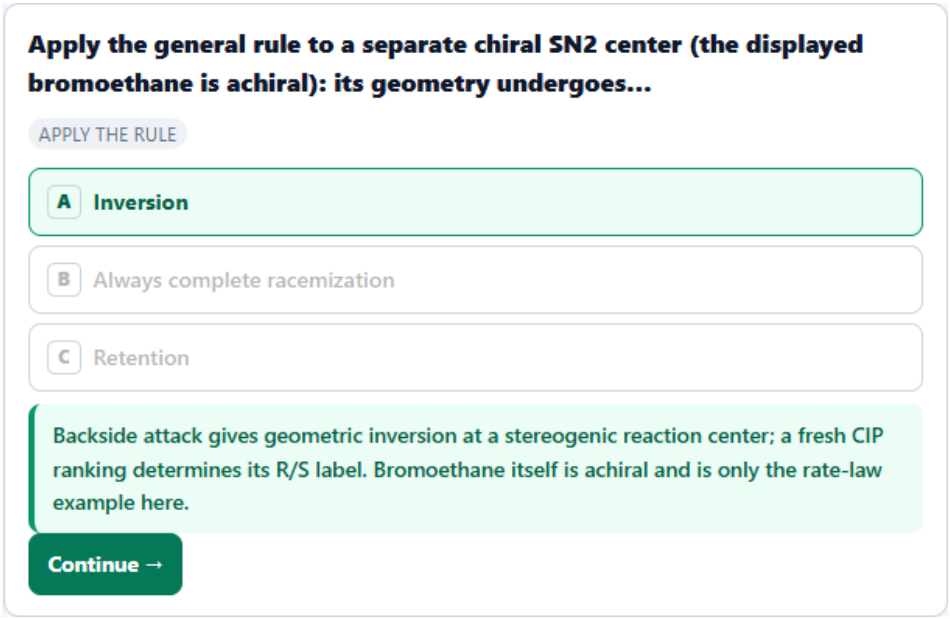

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: The nucleophile approaches opposite the leaving group.

Hint 2: Think of an umbrella turning inside out.

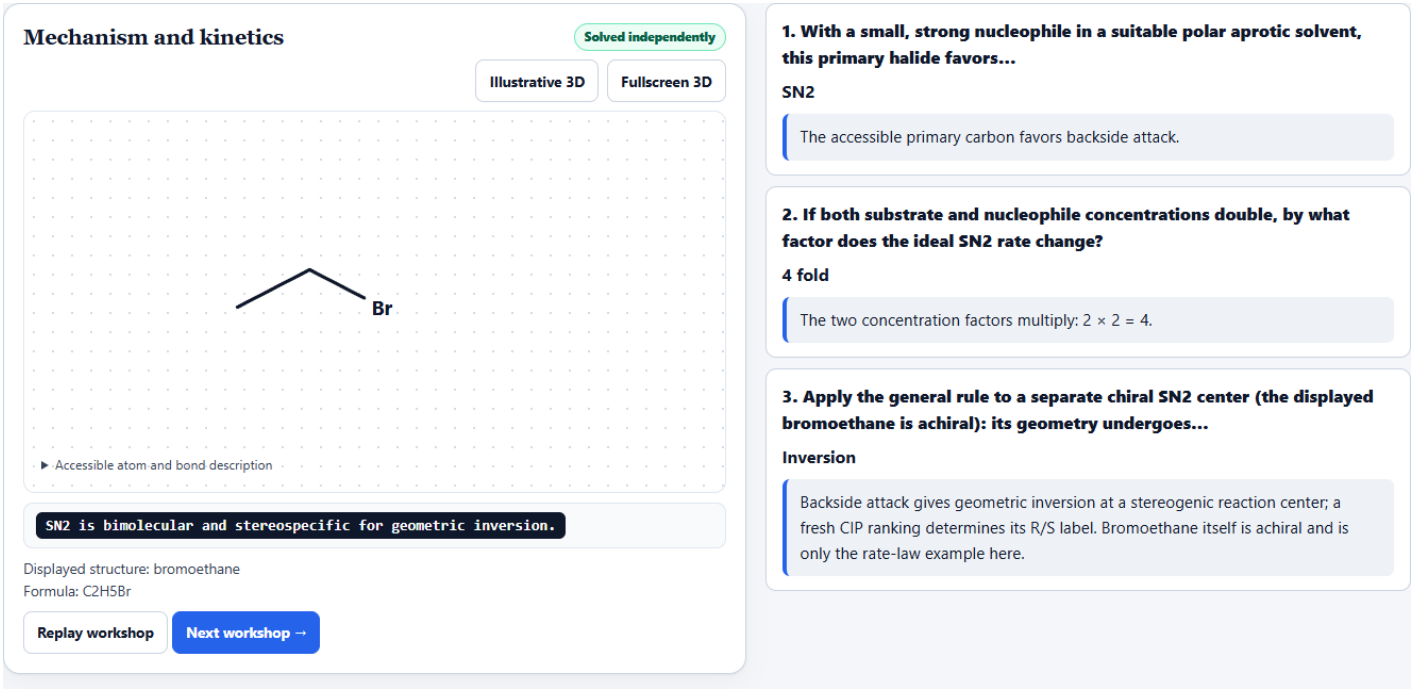
Explanation: Backside attack gives geometric inversion at a stereogenic reaction center; a fresh CIP ranking determines its R/S label. Bromoethane itself is achiral and is only the rate-law example here.

Screenshot: sn2-rate-stage-3-feedback



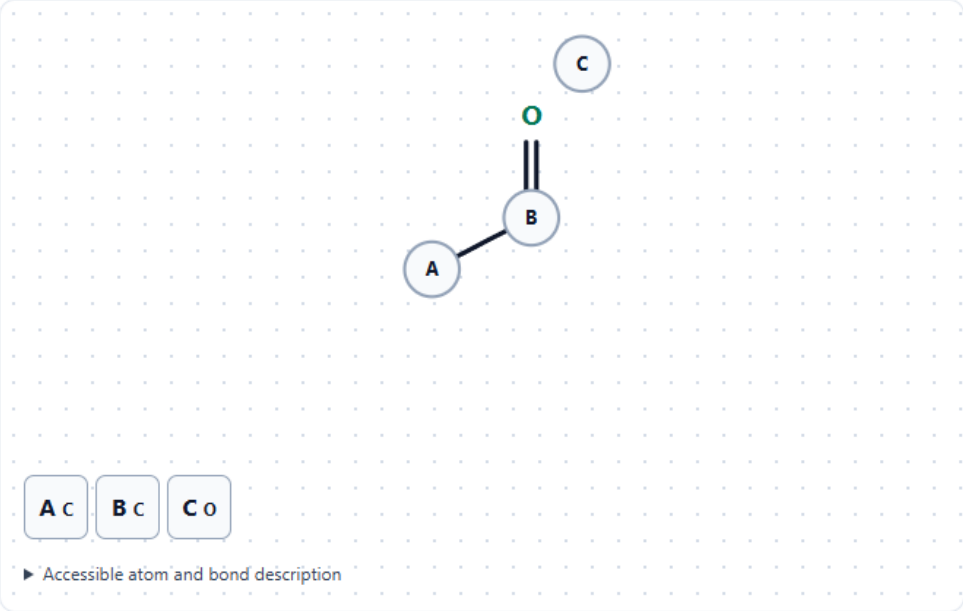

Completion summary: SN2 is bimolecular and stereospecific for geometric inversion.

Screenshot: sn2-rate-complete



# carbon-redox | Assign carbon oxidation state

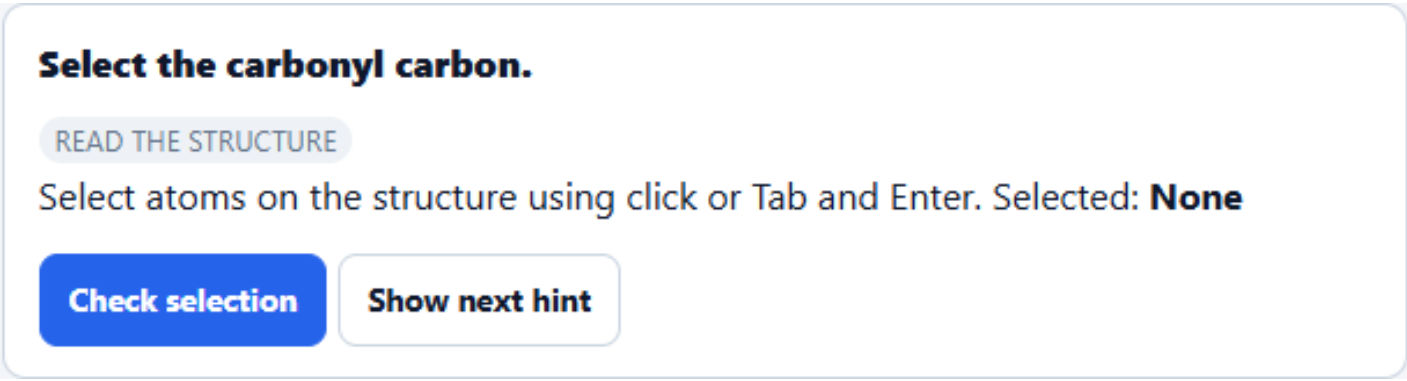
Screenshot: carbon-redox-stage-1-molecule




## carbon-redox:stage-1 | Select the carbonyl carbon.

Answer type: atoms

Screenshot: carbon-redox-stage-1-question



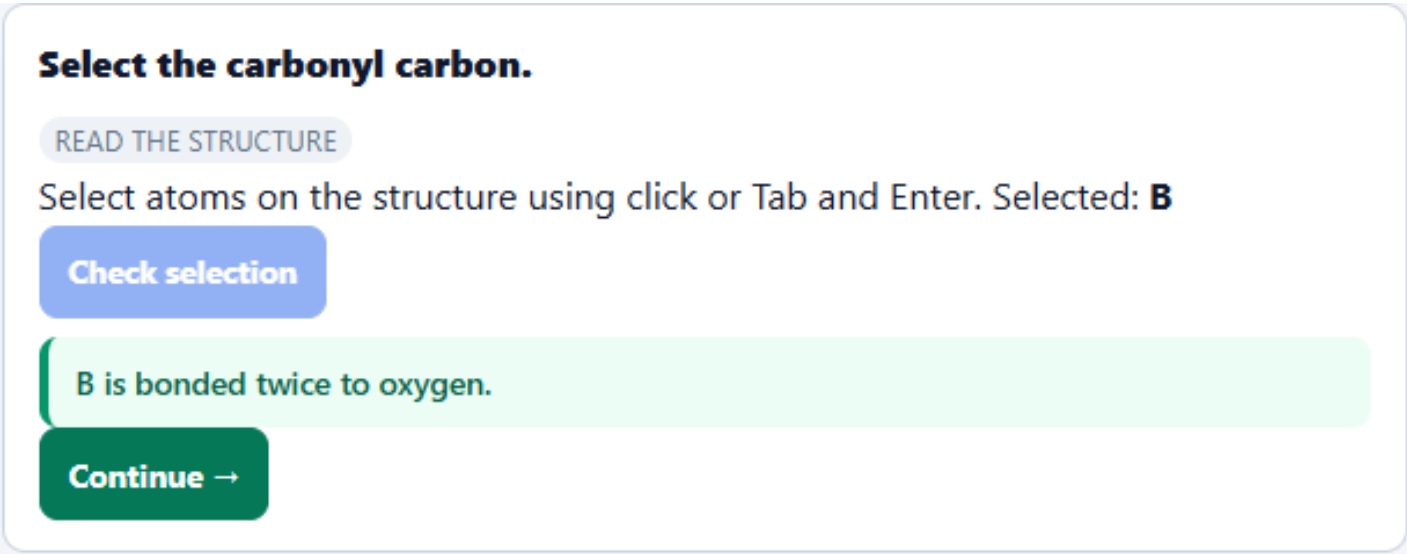

{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1], "sub": [2]}}

Hint 1: Find the C=O bond.

Hint 2: Choose its carbon endpoint.

Explanation: B is bonded twice to oxygen.

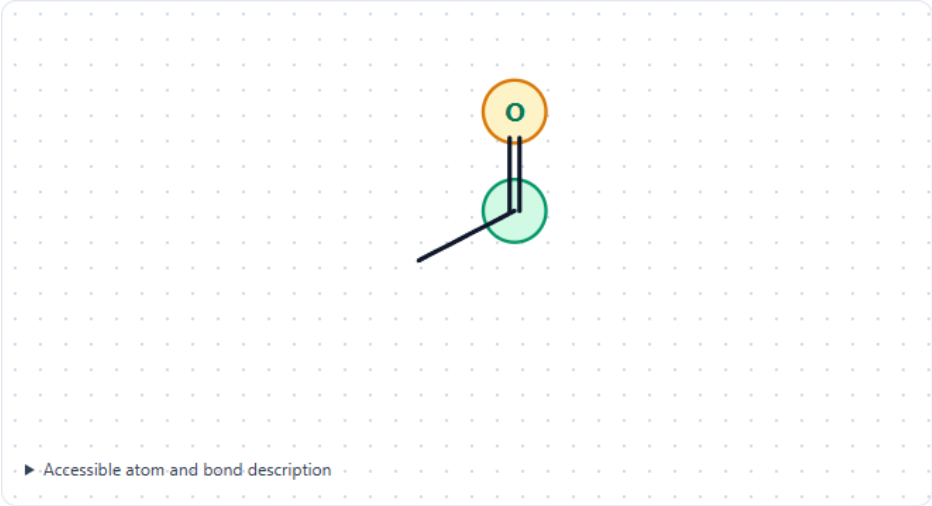
Screenshot: carbon-redox-stage-1-feedback



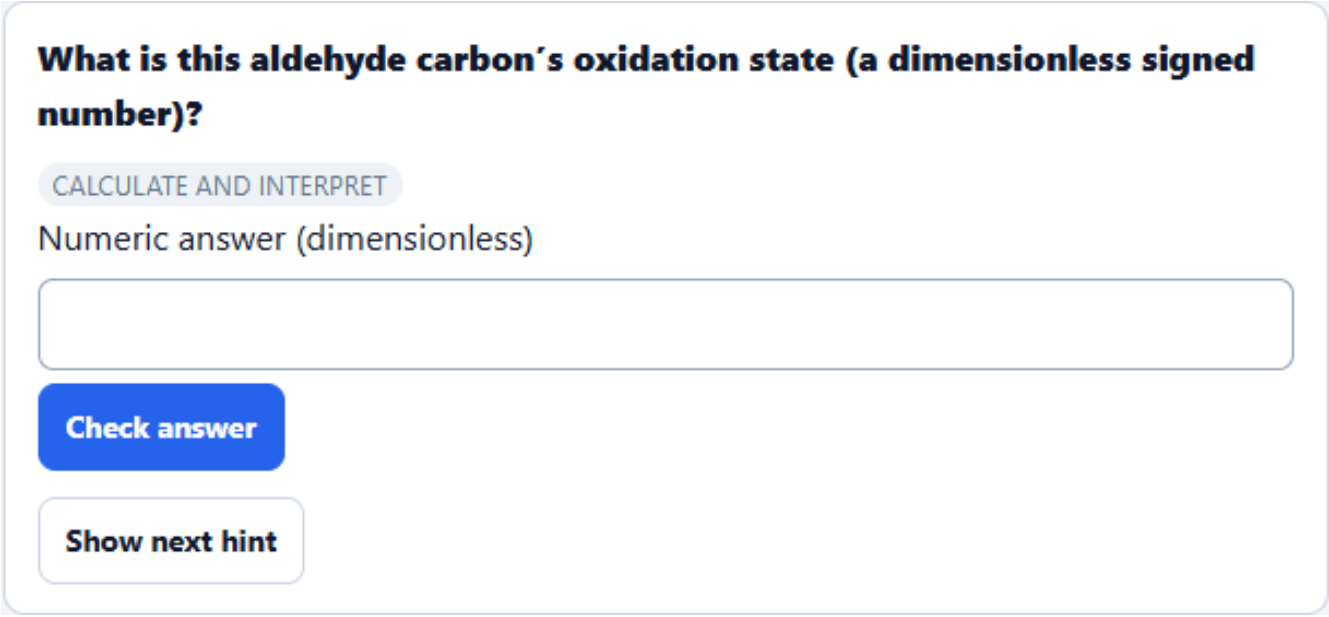

### carbon-redox:stage-2 | What is this aldehyde carbon’s oxidation state (a dimensionless signed number)?

Answer type: number | unit: dimensionless

Screenshot: carbon-redox-stage-2-molecule



Screenshot: carbon-redox-stage-2-question



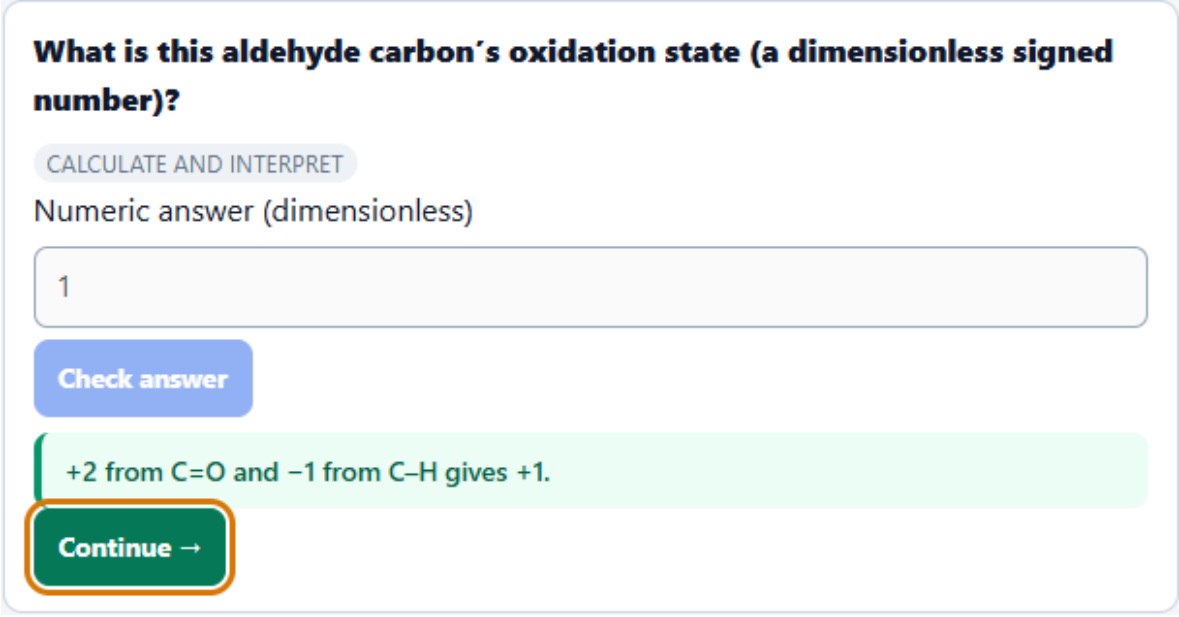

{"authorAnswer": 1, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Assign electrons in C–O to O and in C–H to C.

Hint 2: C–C contributes zero.

Explanation: +2 from C=O and −1 from C–H gives +1.

Screenshot: carbon-redox-stage-2-feedback




## carbon-redox:stage-3 | NaBH4 followed by protonation converts it to...

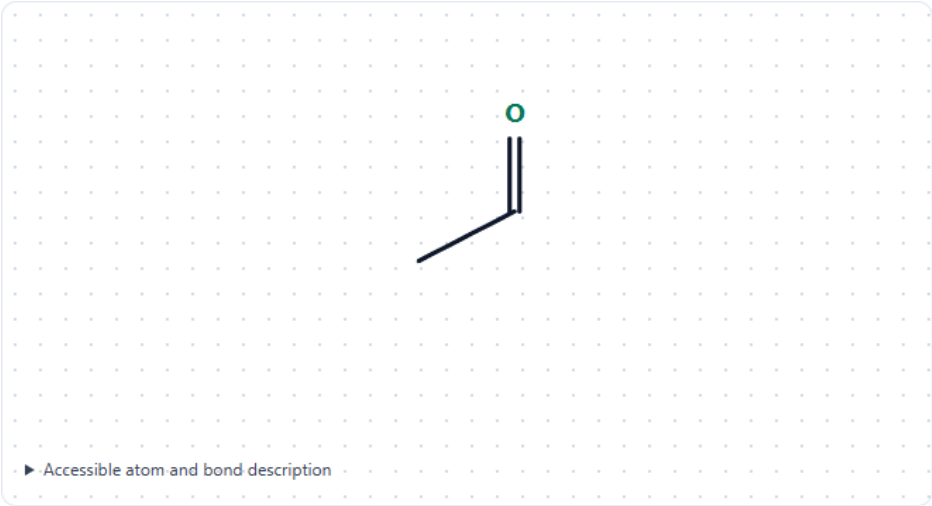
Answer type: choice

Option 0 (A): A primary alcohol

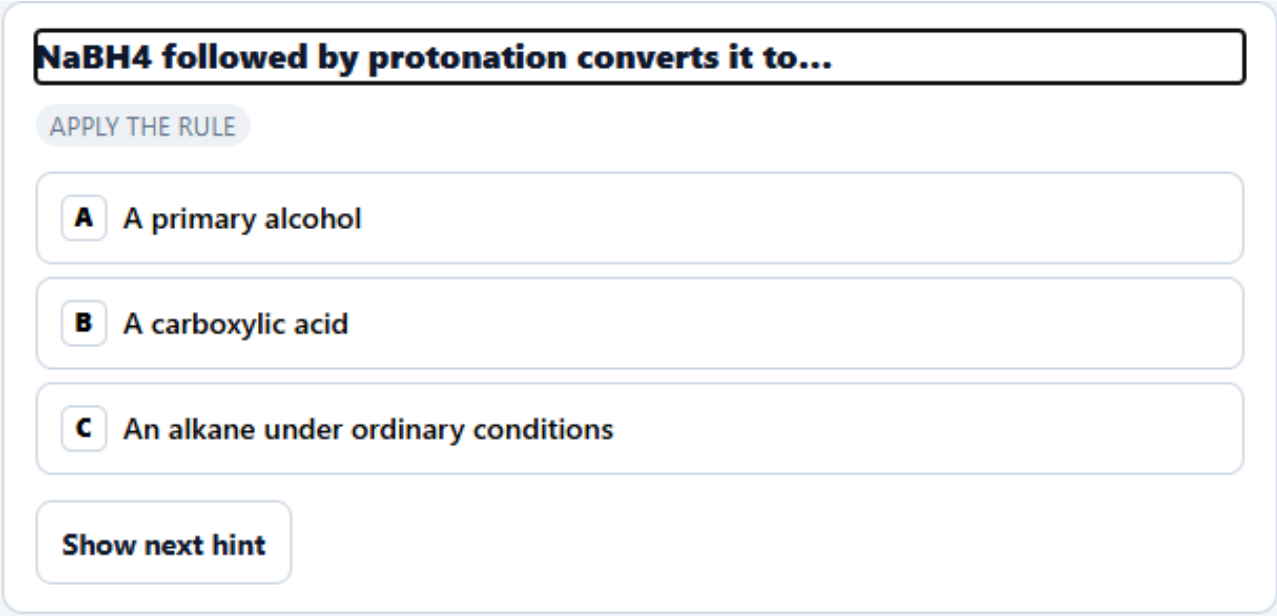
Option 1 (B): A carboxylic acid

Option 2 (C): An alkane under ordinary conditions

Screenshot: carbon-redox-stage-3-molecule




Screenshot: carbon-redox-stage-3-question



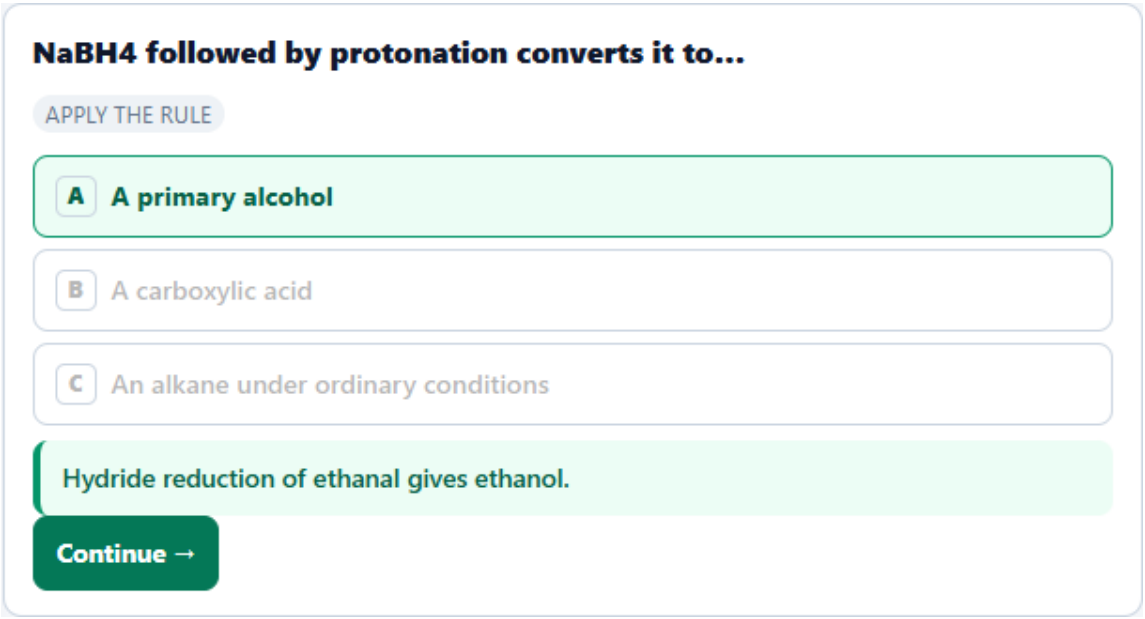

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hydride adds to C=O.

Hint 2: The oxygen remains in the alcohol product.

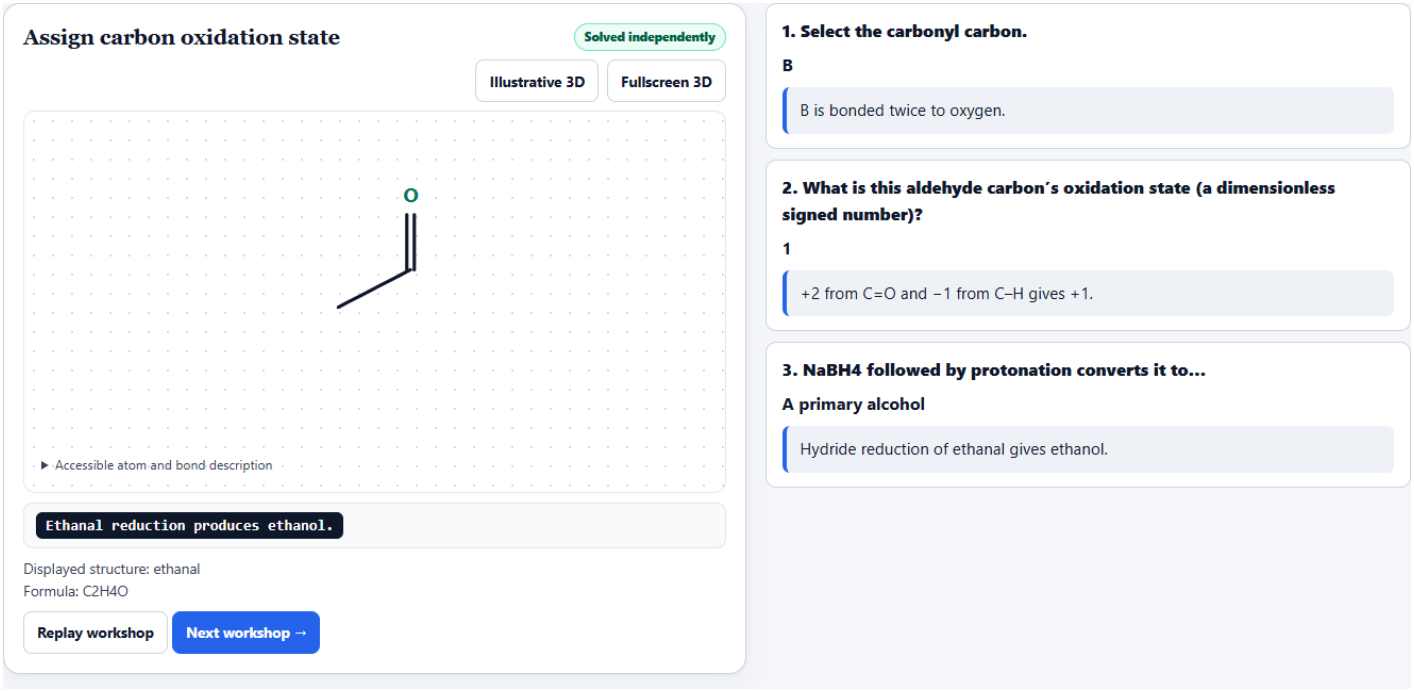
Explanation: Hydride reduction of ethanal gives ethanol.

Screenshot: carbon-redox-stage-3-feedback



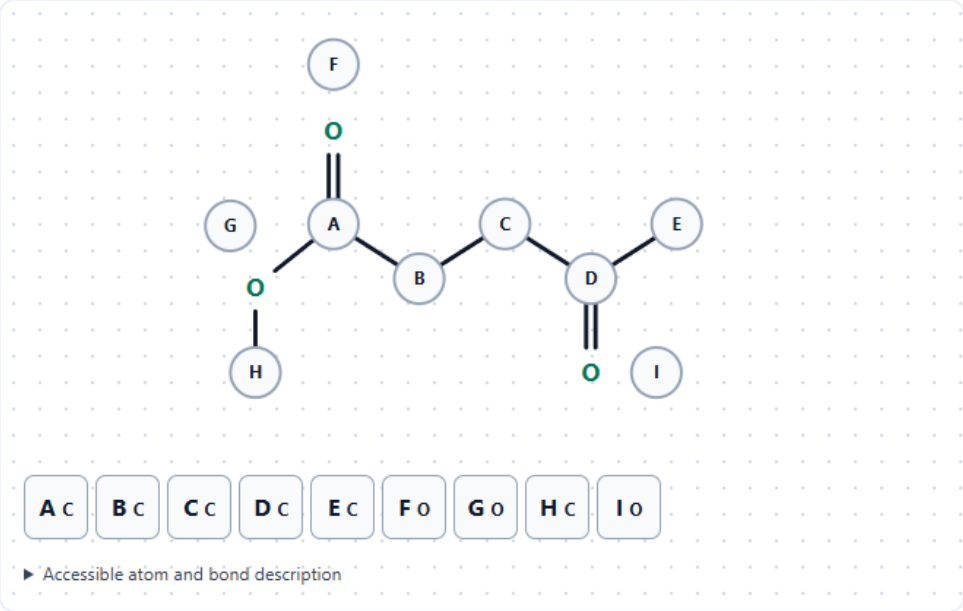

Completion summary: Ethanal reduction produces ethanol.

Screenshot: carbon-redox-complete



# selective-reduction | Select a reactive site

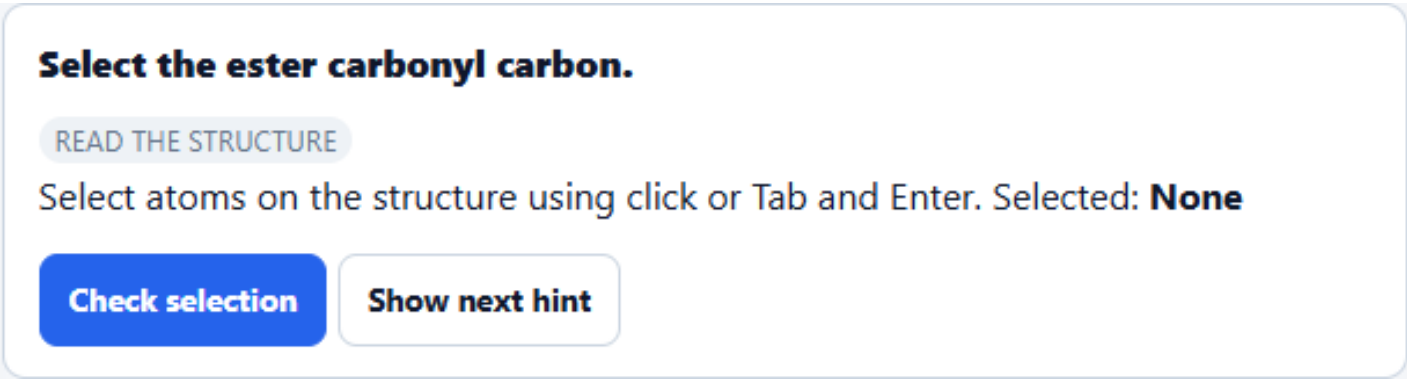
Screenshot: selective-reduction-stage-1-molecule




## selective-reduction:stage-1 | Select the ester carbonyl carbon.

Answer type: atoms

Screenshot: selective-reduction-stage-1-question



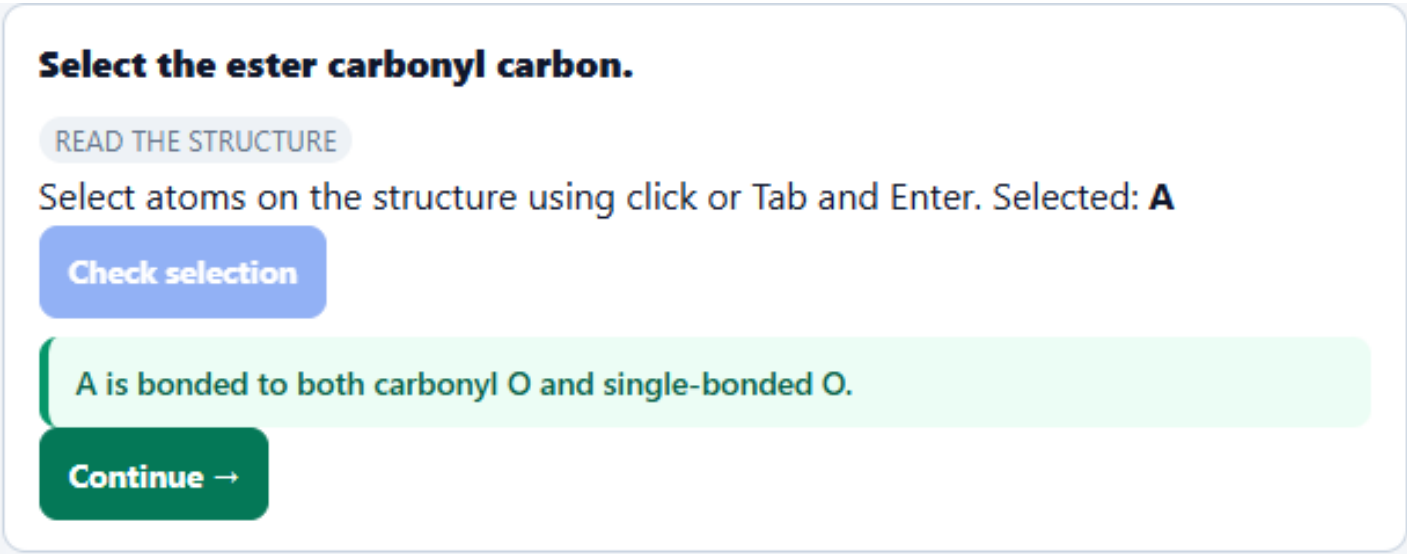

{"authorAnswer": [0], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0, 5, 6]}}

Hint 1: An ester has C(=O)–O–C.

Hint 2: Find the carbonyl with a second oxygen neighbor.

Explanation: A is bonded to both carbonyl O and single-bonded O.

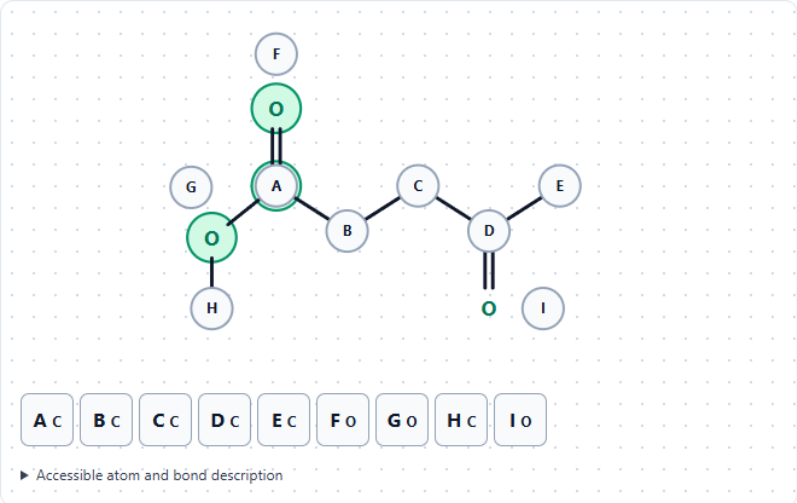
Screenshot: selective-reduction-stage-1-feedback



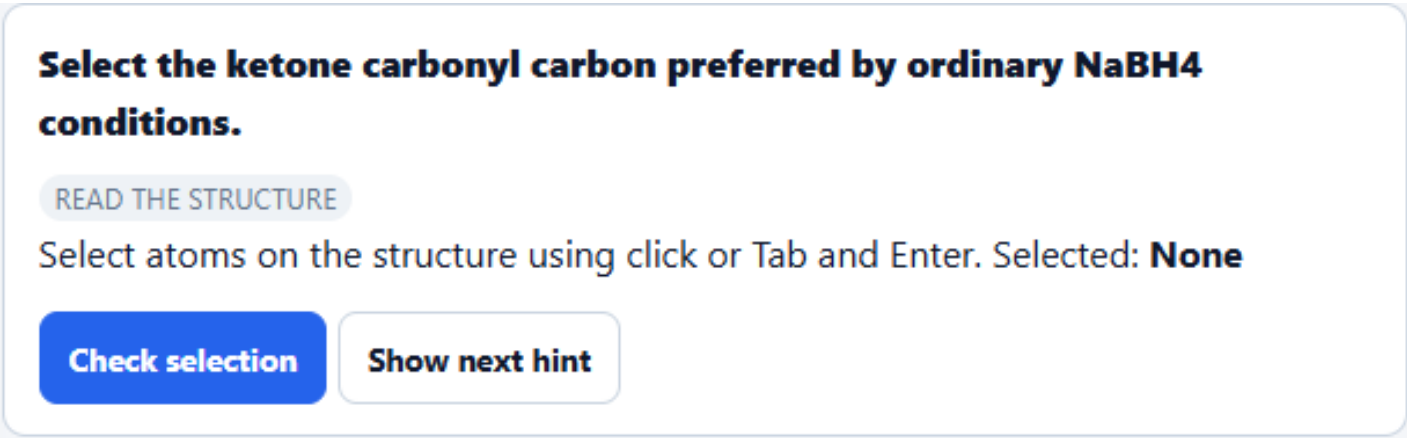

## selective-reduction:stage-2 | Select the ketone carbonyl carbon preferred by ordinary NaBH4 conditions.

Answer type: atoms

Screenshot: selective-reduction-stage-2-molecule



Screenshot: selective-reduction-stage-2-question



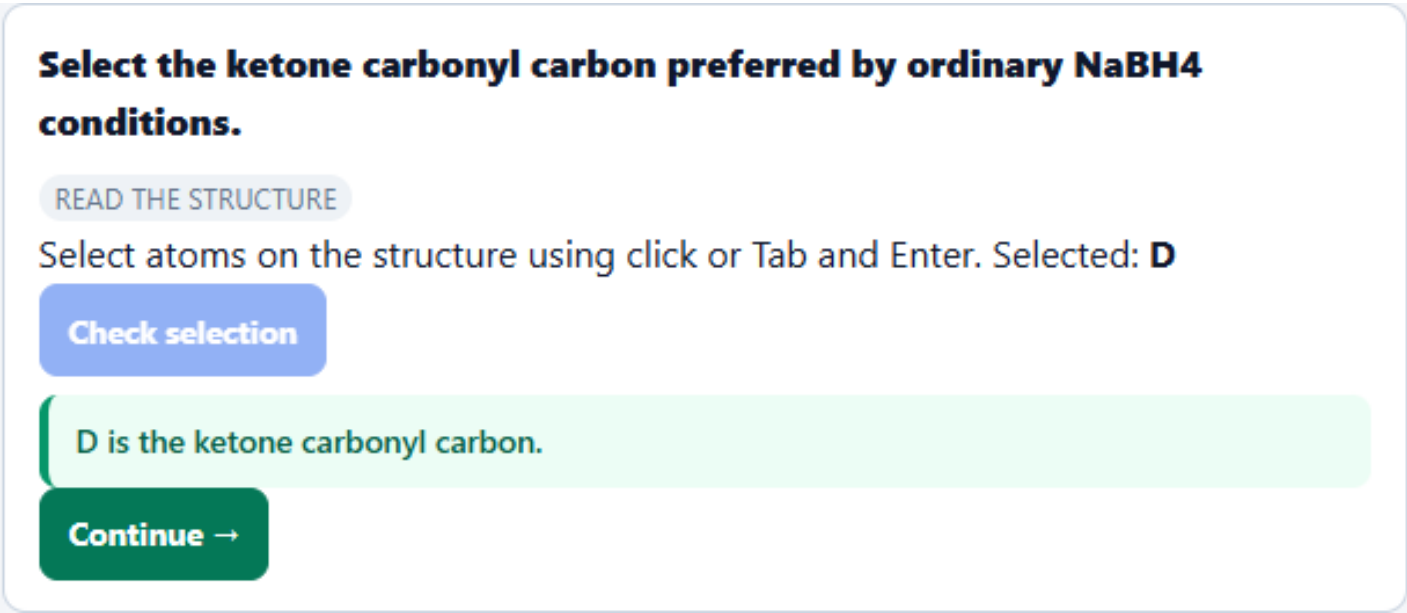

{"authorAnswer": [3], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [3, 8]}}

Hint 1: The ketone has carbon substituents on both sides.

Hint 2: The right-hand carbonyl is more readily reduced here.

Explanation: D is the ketone carbonyl carbon.

Screenshot: selective-reduction-stage-2-feedback




### selective-reduction:stage-3 | Why not simply select the highest-priority naming group?

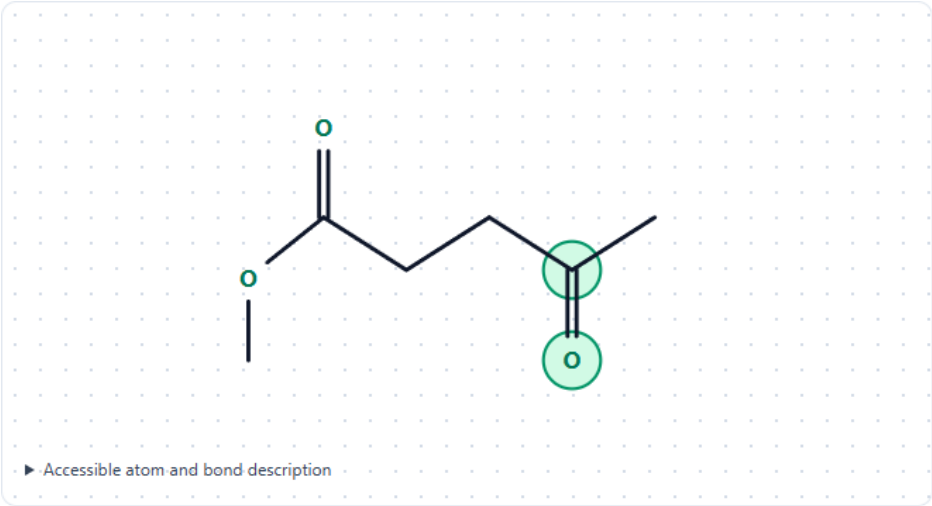
Answer type: choice

Option 0 (A): Naming seniority does not determine kinetic reactivity

Option 1 (B): The ester has no carbonyl

Option 2 (C): All carbonyls react identically

Screenshot: selective-reduction-stage-3-molecule




Screenshot: selective-reduction-stage-3-question

**Why not simply select the highest-priority naming group?**

APPLY THE RULE

**A** Naming seniority does not determine kinetic reactivity

**B** The ester has no carbonyl

**C** All carbonyls react identically

**Show next hint**

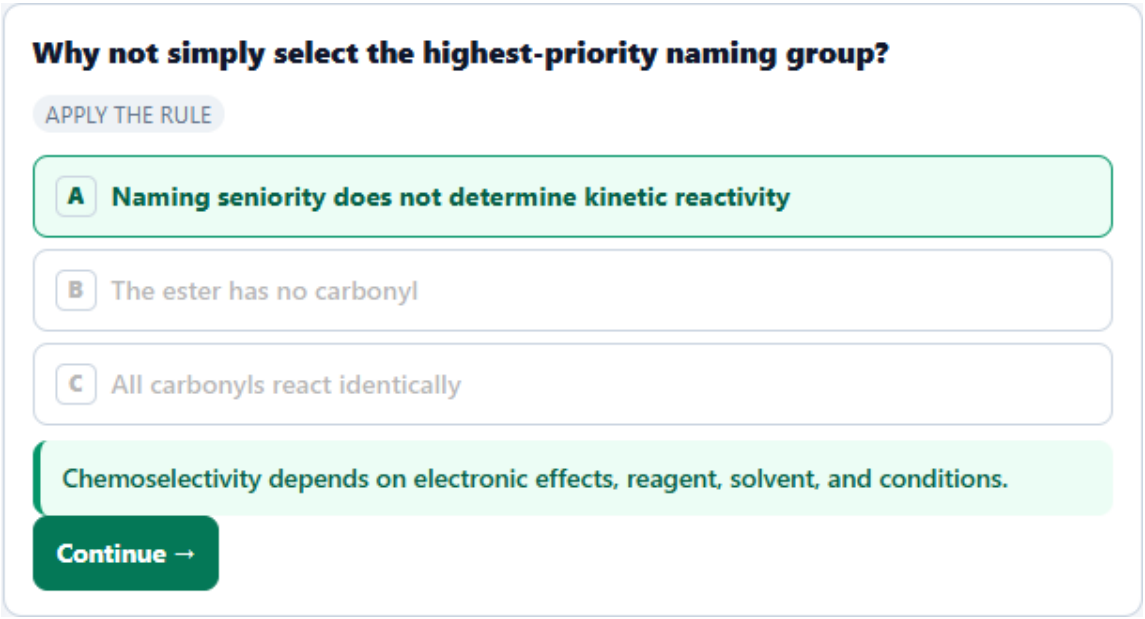
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Naming and reaction mechanisms solve different problems.

Hint 2: Compare resonance donation and electrophilicity.

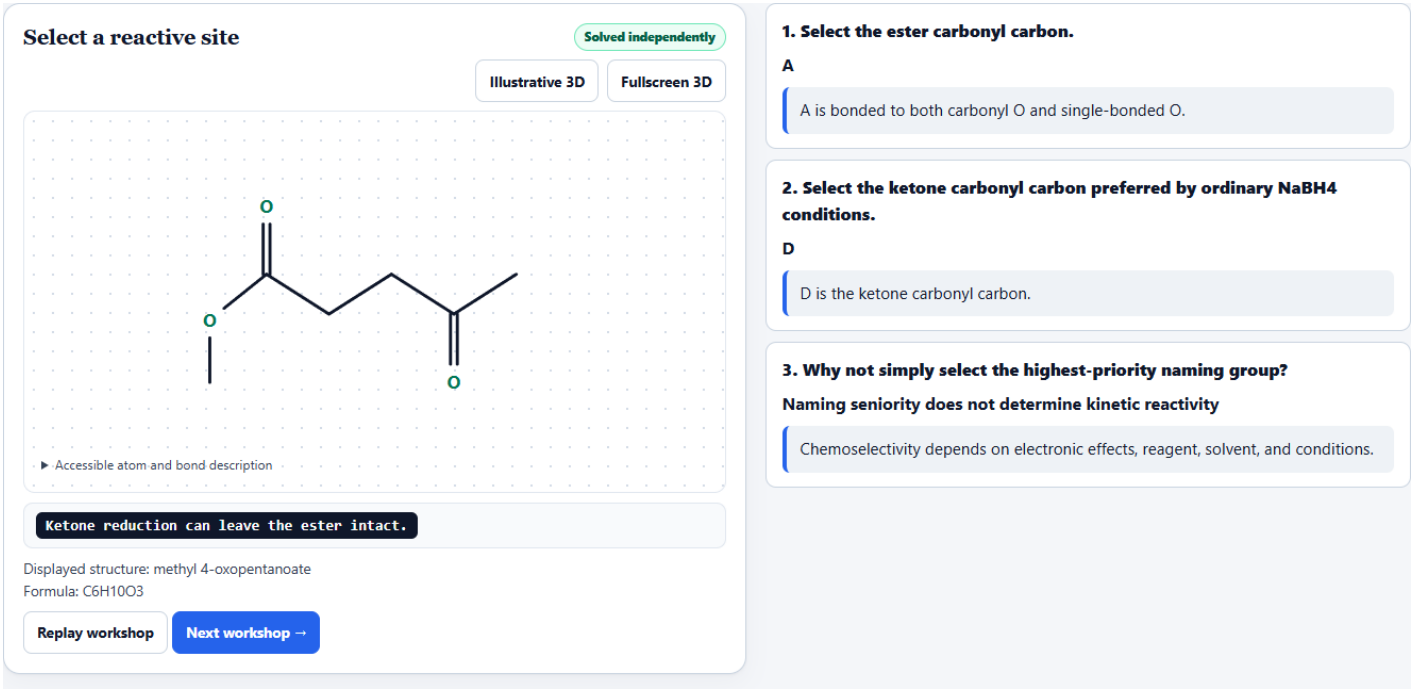
Explanation: Chemoselectivity depends on electronic effects, reagent, solvent, and conditions.

Screenshot: selective-reduction-stage-3-feedback




Completion summary: Ketone reduction can leave the ester intact.

Screenshot: selective-reduction-complete



# Molecular repository: author claims to verify

## aceticacid

Author claims: ethanoic acid | formula C2H4O2 | canonical composition C2H4O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=108.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

## acetate

Author claims: acetate (one resonance contributor) | formula C2H3O2 | canonical composition C2H3O2 | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=175.00, y=115.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

## acetaldehyde

Author claims: ethanal | formula C2H4O | canonical composition C2H4O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=131.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=189.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=189.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2]]

## bromoethane

Author claims: bromoethane | formula C2H5Br | canonical composition C2H5Br | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): Br; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

## tbutylbromide

Author claims: 2-bromo-2-methylpropane | formula C4H9Br | canonical composition C4H9Br | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=85.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=168.00, y=85.00; charge=0; explicit H override=none

3 (D): C; x=108.00, y=25.00; charge=0; explicit H override=none

4 (E): Br; x=108.00, y=145.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1]]

## ethanol

Author claims: ethanol | formula C2H6O | canonical composition C2H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

## acetone

Author claims: propanone | formula C3H6O | canonical composition C3H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

## ketoester

Author claims: methyl 4-oxopentanoate | formula C6H10O3 | canonical composition C6H10O3 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=70.00, y=100.00; charge=0; explicit H override=none

1 (B): C; x=125.00, y=135.00; charge=0; explicit H override=none

2 (C): C; x=180.00, y=100.00; charge=0; explicit H override=none

3 (D): C; x=235.00, y=135.00; charge=0; explicit H override=none

4 (E): C; x=290.00, y=100.00; charge=0; explicit H override=none

5 (F): O; x=70.00, y=40.00; charge=0; explicit H override=none

6 (G): O; x=20.00, y=140.00; charge=0; explicit H override=none

7 (H): C; x=20.00, y=195.00; charge=0; explicit H override=none

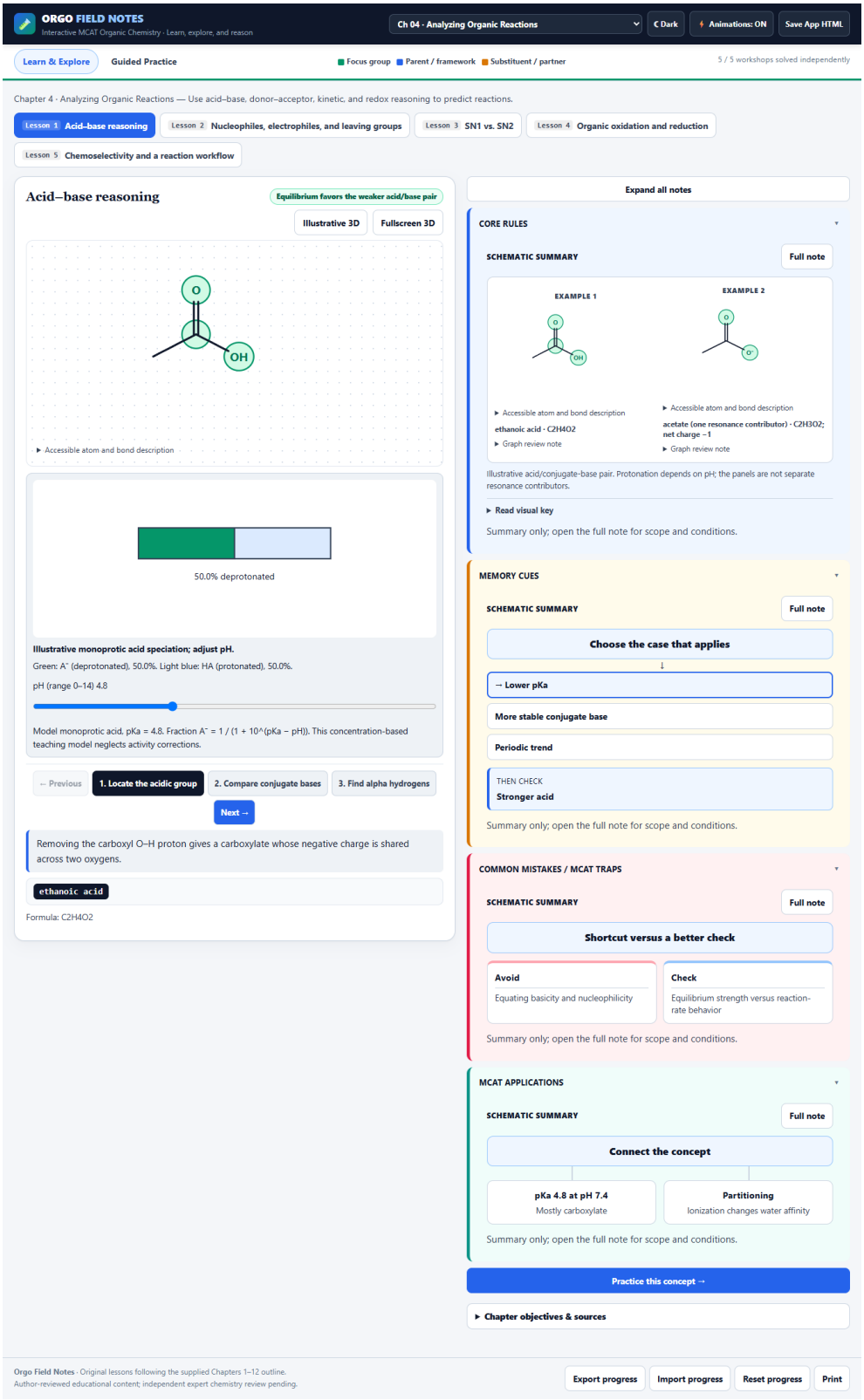
8 (I): O; x=235.00, y=195.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [0, 5, 2], [0, 6, 1], [6, 7, 1], [3, 8, 2]]

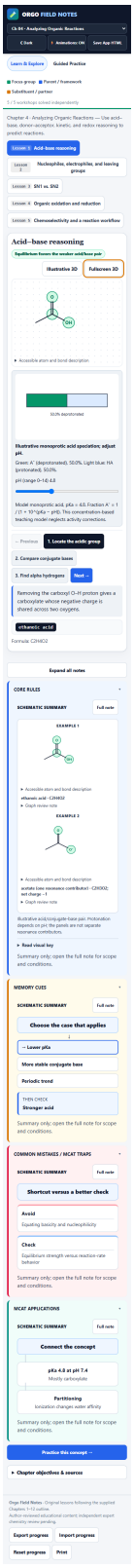
# Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

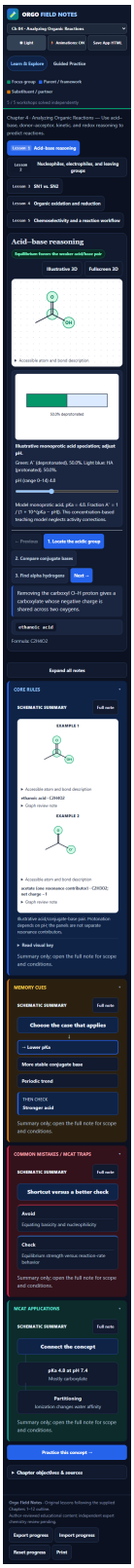
Screenshot: layout-desktop-light



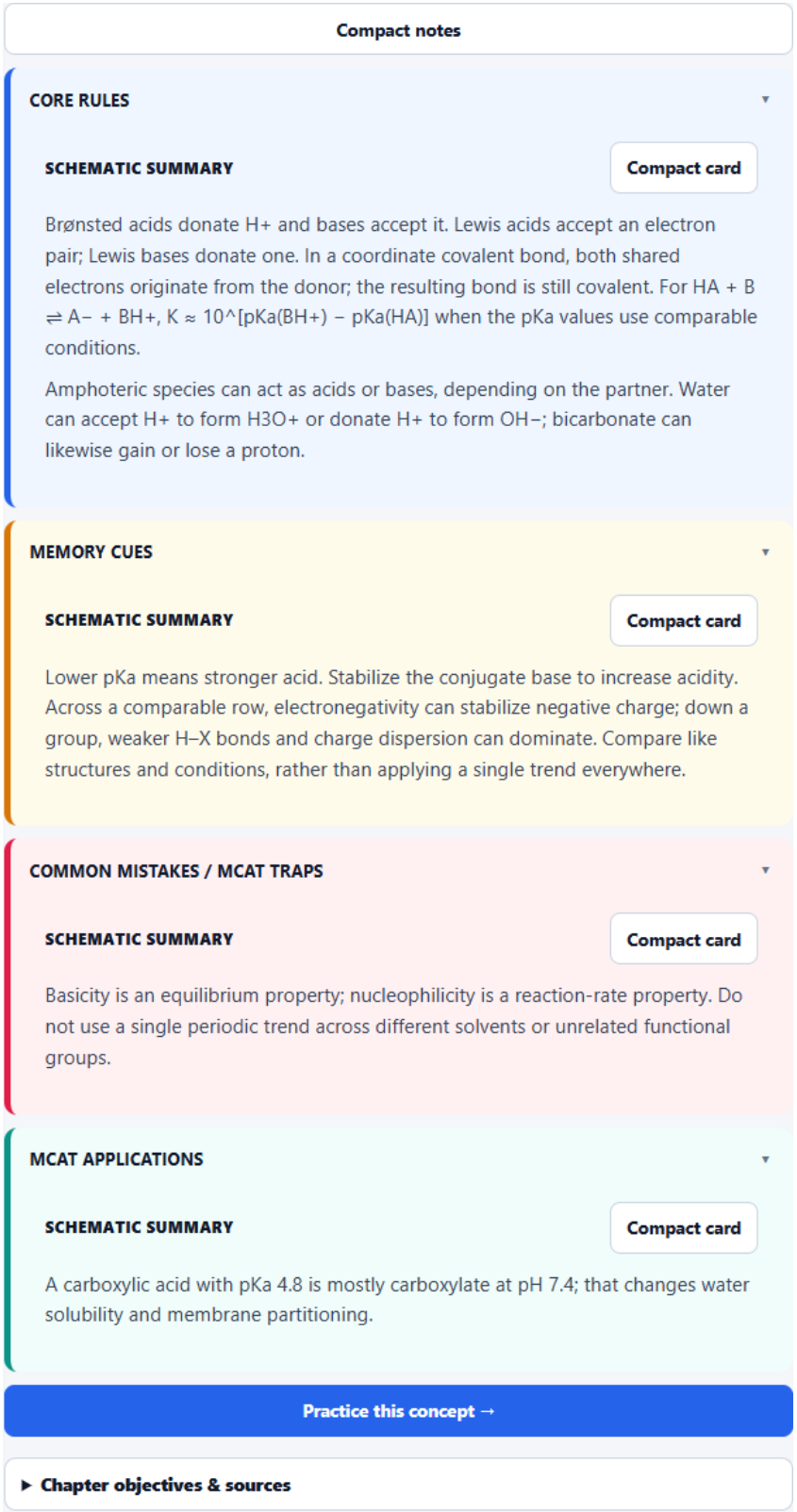
Screenshot: layout-mobile-light



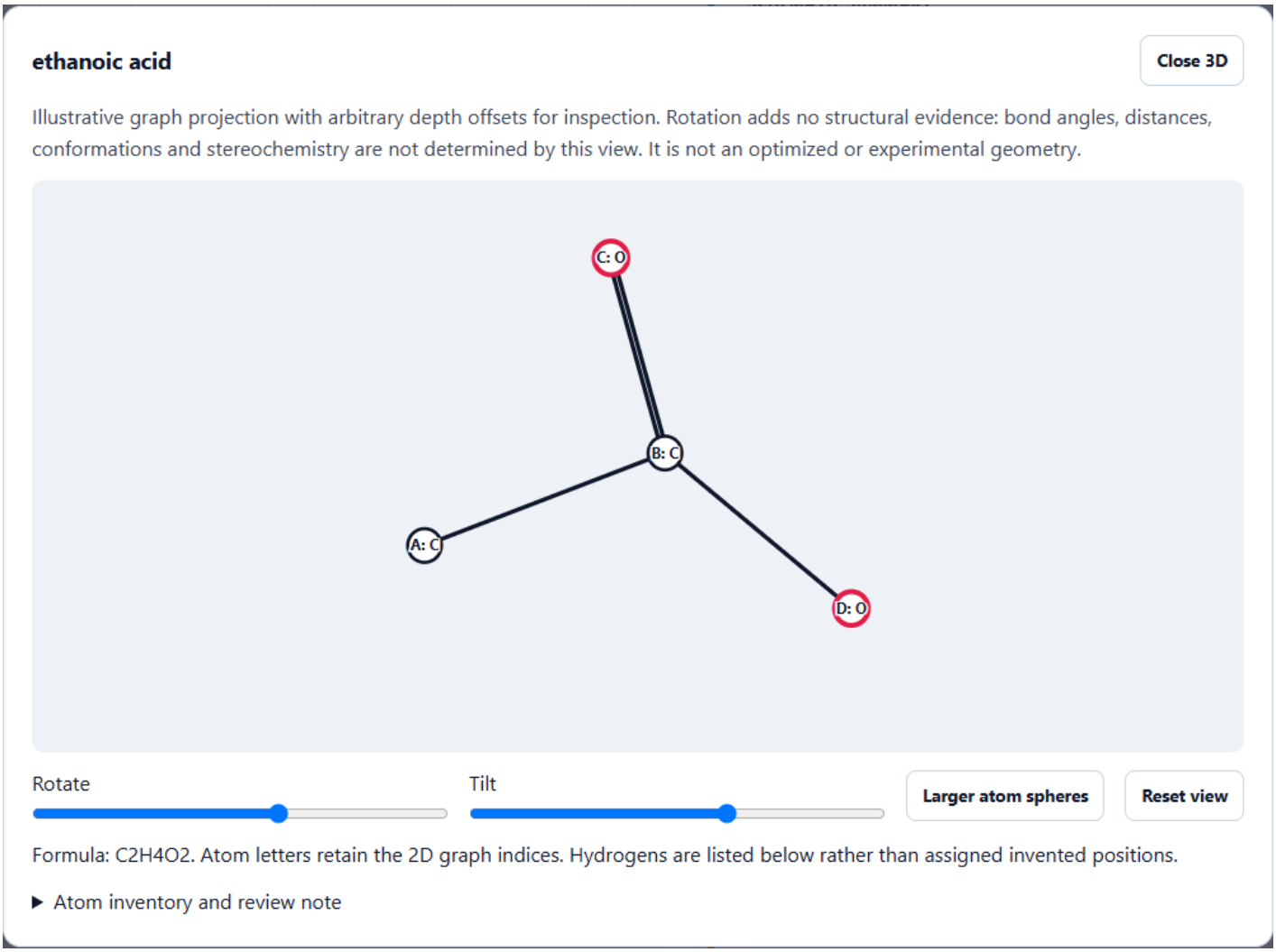
Screenshot: layout-mobile-dark



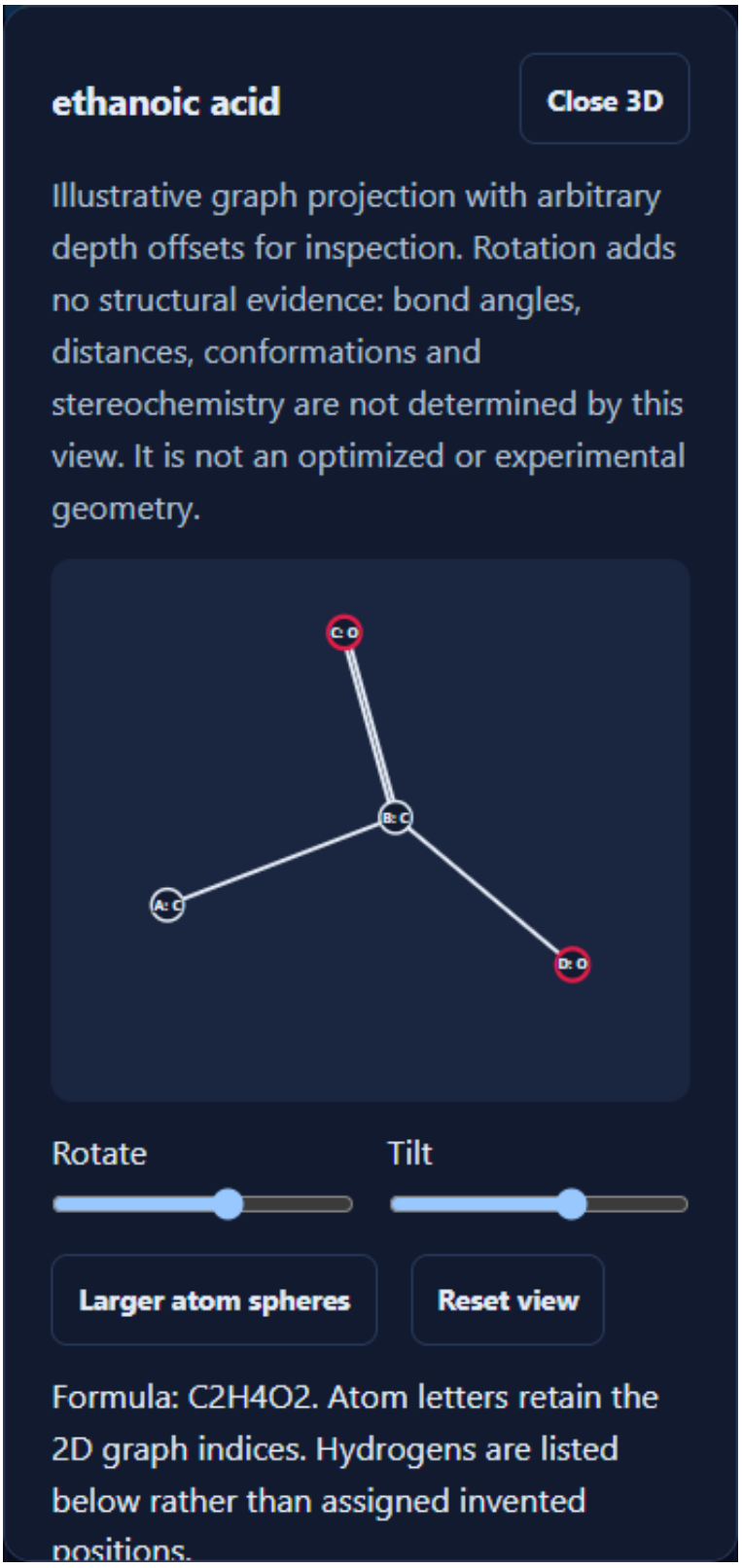
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark




# Execution evidence

77 screenshots and 18 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.